

Final Thesis Report

Optimizing the Early Detection of Dementia by Tracking the Progression of Parkinson's Disease Using Deep Learning and Computer Vision

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A thesis submitted to the Department of Computer Science and Engineering
in partial fulfillment of the requirements for the degree of
B.Sc. in Computer Science and Engineering

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Declaration

It is hereby declared that

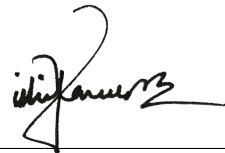
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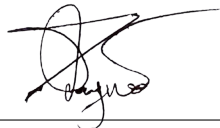
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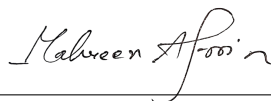
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Abstract

Dementia is a future complication in Parkinson’s disease patients 40-80% of the time. Aggregate proteins called Lewy Bodies propagate in the brain as Parkinson’s becomes increasingly critical, and once they damage the substantia nigra and break into the cortex, dementia is induced. Purely clinical assessments greatly decrease the treatment quality for Parkinson’s patients because neurologists rely heavily on several varying indicators over time to definitively diagnose the disease. Incorporation of deep learning models significantly improves the accuracy of detecting if and when Parkinson’s progresses to dementia. In this paper, we use several deep learning models to establish a comprehensive link between Parkinson’s disease and dementia, which works by first optimally detecting Parkinson’s disease in patients, and then, using multiple weighted MRI modalities, transfer learning, and multimodal fusion to merge imaging modalities with clinical features, as well as conducting ablation studies, in order to determine the most efficient configuration to detect the presence and degree of cognitive impairment which manifest as dementia in Parkinson’s patients. Preprocessing and extracting suitable features from MRI scans and multiple-visit temporal clinical features obtained from both the Parkinson’s Progression Markers Initiatives (PPMI) dataset and a limited subset of this dataset, we use our baseline single-modality and multi-modal framework to delineate whether a patient will develop dementia by how our system predicts cognitive decline, and incorporate Attention-Augmented explainability to find which brain regions play the biggest role in signaling decline. Using our study, we aim to obtain results with enhanced accuracy that show considerable speeding up of the time around which dementia induced by Parkinson’s is typically detected and diagnosed, thereby reducing long-term healthcare costs by starting to plan and enhance treatment early on, improving drug trials, and improving the quality of life of patients.

Keywords: Parkinson’s disease, Early detection, Dementia, Cognitive Decline, MRI scans, Temporal clinical features, Transfer learning, Ablation study, Preprocessing, Feature extraction, Parkinson’s Progression Markers Initiatives (PPMI), Multi-modality, Deep learning, Attention-Augmented Explainability.

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Chapter 1

Introduction

1.1 Background

Parkinson's disease (PD) is a neurodegenerative disease that manifests in the form of motor disorder and impairment in cognitive function. Its primary motor symptoms fester as bradykinesia or slowness in movement, resting tremors, and physical rigidity. Later, instability in posture and balance, dyskinesia or involuntary jerking movements, dystonias or painful muscle contractions, and dysfunctional gait appear [1], as well as non-motor manifestations such as sensory issues, sleep disorders, anxiety disorders, and dementia [2]. Dementia is an umbrella term for an amalgamation of disorders and diseases that extend cognitive impairment to areas such as judgement, thinking, memory, coordination, etc., causing striking changes in personality and rendering patients confused, forgetful, disoriented, withdrawn, and anxious [3]. The number of dementia cases is expected to triple in the next twenty-five years, as the global population ages, resulting in a total of 152 million [4]. Based on existing studies regarding the relationship between PD and dementia, we surmise that two main kinds of dementia occur due to the progression of PD primarily in older patients over the age of 65 [3]: Lewy Body Dementia (LBD) and Parkinson's Disease Dementia (PDD) [5]. Studies show [6] that early lifestyle maintenance and regulation of risk factors are helpful in reducing the risk of dementia in about 40% cases.

Tests and scales that measure the health of cognitive functions are typically used in clinical settings to diagnose dementia, administered by specialist doctors. Some of these tests include the Mini-Mental State Examination (MMSE), which was first introduced in 1975 and is a 6 to 10 minute test that assesses non-motor functions such as language command, attention, memory, orientation, and visuospatial coordination with a total score of 30 [7]; the Addenbrooke's Cognitive Examination (ACE) and ACE-Revised (ACE-R), which respectively assess fluency, attention, memory, language, and then these same functions more extensively with added visuospatial functions, conducted in 15 to 20 minutes with a score of 100 [8]; the Montreal Cognitive Assessment (MoCA), which is a 10-minute test that assesses out of 30 the executive functions, attention, language, memory loss, and fluency in language and phonemics of a patient [9]; and the Clock Drawing Test (CDT), which is a seemingly easy and very fast test that requires the subject to draw a clock, including all the hours, and point its hands to a specified point of time, such as fifteen to four. The

subject’s spatio-visual and coordinative executive functions are evaluated based on the correctness of the overall drawing [10]. To track and understand the progression of PD, the Unified Parkinson’s Disease Rating Scale distributed by the Movement Disorder Society (MDS-UPDRS) is used [11], with a total of 50 questions divided into four parts to gauge both motor and non-motor experiences faced by the patients in their activities of daily living (ADL).

But clinical scales have far too many limitations to be hailed as all-encompassing and sufficient in the diagnosis of dementia, such as the MMSE’s inability to present results with both high sensitivity and specificity [7], and if that problem is resolved [8][9], then there remain other shortcomings, like the over-dependence on the presence of trained clinicians to conduct and interpret the tests, the time needed to complete and evaluate the tests, and the loss of vital subjective clinical information - such as the speed of drawings, pauses, hesitations, straining to recall things - that goes undocumented because of the manual aspect of the assessments [12]. This is where the need comes in for automated models that carry out these tests by incorporating not only test performances but also the cognitive behaviors of subjects; calculate their results and provide diagnoses and predictions with better accuracy, in a time-space-cost-optimized manner. Some of the aforementioned clinical tests, like CDTs, have been automated using machine learning (ML) techniques - for example, in a paper, subjects were assessed by their drawing of a clock on a digitizing tablet in an Attentional Matrices Test (AMT) [13], and their performance was evaluated by ensembling the results of four ML models: logistic regression, K-nearest neighbors (KNN), random forests (RF), and support vector machine (SVM) to classify whether or not the participants had dementia or were a part of the healthy control group.

Besides automating just clinical assessment scales, there have been multiple attempts, both successful and unsuccessful, to shift the responsibility of disease detection, progression, and stage prediction to AI, ML, and DL models because of their effectiveness, swiftness, and novel benefits. For example, from medical records that document dexterity and level of control of subjects’ hands and fingers, features are extracted using magnetic sensors by converting their movements into wave signals; these features are passed through a one-class SVM to classify subjects with dementia from those who are healthy [14]. Skin color segmentations, OpenPose Skeleton Model (a DL procedure), and linear discriminant analysis can also be utilized to achieve this end with different performance rates. Recently, eye movement assessments have become a popular non-invasive test that aids dementia’s early detection. By measuring the eye movements of test subjects while they quietly read, using an EyeLink 1000 Desktop Mount and other devices, important features are extracted and passed onto ML models like the Naive Bayes classifier, SVM, etc. to differentiate between healthy controls, people with mild cognitive impairment (MCI), and people with subjective cognitive impairment (SCI) [15][16].

Besides movements of the limbs and eyes, acoustic or sound features also come handy in the detection of dementia. Speech tasks given to test participants can be used to extract features and capture segmented information using techniques like active data representation (ADR) [17]. Natural language processing (NLP)

techniques are also useful, such as word2vector and GloVe, to derive features from the conversations between subjects [18], and linguistic data extraction strategies like bag-of-words (BOW) and term frequency inverse document frequency (TF-IDF) can also be used in a multimodal architecture, by combining the extracted data with non-linguistic data based off clinical evaluations, like blood tests, cerebrospinal fluid conditions, and other important biomarkers [19]. Deep learning models have consistently given results worth noting in detecting dementia by capturing small changes in image scans even at the earliest phase of the disease. CNN models with several deep convolutional layers have been useful in differentiating between people that have Alzheimer’s Disease (AD), or MCI, or those who are healthy subjects, using features preprocessed and extracted from 3D MRI scan data [20]. Data obtained from PET scans can similarly be used [21]. For PD detection, Feed-forward Neural Networks (FNN) [22] and Deep Belief Networks (DBN) using unsupervised learning assisted by Restricted Boltzmann Machines (RBM) for optimization [23] have shown considerable results.

Evidence from papers have found that the discriminative neuroanatomical biomarkers that contribute most prominently to detecting dementia during the progression of PD are volumetric changes, particularly of the lateral ventricles along with the hippocampus, and the reduction of white matter [24]. Procedures have been demonstrated using Multivariate Filter-Based Naive Bayes and Filter Selection Naive Bayes to observe the progression of PD into the development of dementia [25]. Hybrid models like the LightGBM-TabPFN [24] have been used quite effectively to predict an approximate time wherein a PD patient may show signs of dementia, with this model being efficient in dealing with missing values and also being useful when it comes to small datasets. The model’s elevated interpretability using SHAP analysis (SHapley Additive exPlanations) made its results more accessible and helpful to clinicians.

Even after so many attempts to introduce automated methods to measure Parkinson’s progression and predict dementia detection, these two separate fields still largely rely on cognitive tests which are both inefficient and time-consuming. The integrated realm of study connecting PD and dementia, within which their interrelation is investigated, and where we try to predict the occurrence of one due to the progression of the other, is still quite limited - but it has vast potential for exploration. Although certain body fluid biomarkers and brain scans are able to track PD at each stage and detect dementia in its early stages, the high cost and invasiveness of such procedures make it impractical for widespread use [6]. Using deep learning and advanced machine learning, we can identify these biomarkers from image scans or processed tabular data, and use them to determine the probability of co-occurrence of the two diseases, specifically the probability of development of dementia over the years of Parkinson’s progression.

1.2 Rationale of the Study or Motivation

Parkinson’s Disease (PD) and Dementia are two neurodegenerative diseases that affect a vast amount of people of the world for which the affected patients struggle socially, economically and in many other manners. Through research it is found that the progression of Parkinson’s Disease is likely to result in Dementia. But detecting this at an early stage could enhance the patient’s condition by taking necessary medical assists. Though there are now many advanced tools and techniques, determining the detection of Dementia in PD patients still seems to be difficult enough as the symptoms and natures are less likely to be discovered.

Through this research we intend to find out the gap where by utilizing the proper technological advancements and with the help of artificial intelligence like deep learning and computer vision, we can detect the transition from Parkinson’s Disease to Dementia at an early stage by keeping track of cognitive decline. With the help of deep learning algorithms we can process and analyze a huge amount of datasets and identify the patterns for the detection which often seems to be overlooked in the traditional methods. Moreover, through this approach we can find better health-care solutions with the increasing population, and in cost effective ways. Through this study we wish to merge these advanced algorithms and techniques with medical expertise, as well as to merge imaging modalities with clinical features, for a better result of detecting dementia at an early stage in confirmed PD patients.

By analyzing past research works, we found out gaps in them which needs to be addressed for the betterment of our research:

1. **Lack of Suitable Datasets:** Datasets containing exact features that correlate to both PD and dementia aren’t widely available.
2. **Lack of Combined Frameworks:** Most existing research regarding these two diseases isn’t integrated—they concentrate on each disease respectively. Their co-occurrence and correlation, especially using multi-visit PD data to detect cognitive decline which transitions eventually to dementia, have not been extensively studied.
3. **Lack of Dynamic Assessments:** Dynamic assessments of disease progression using time-series data are still uncommon, as most research focuses on static detection.
4. **Lack of AI Explainability:** Current AI-DL models and their predictions/results aren’t easy for clinicians to interpret reliably.
5. **Lack of Generalized Results:** Existing biases in models and the usage of homogeneous and small datasets lead to results that are not applicable to larger, diverse populations.
6. **Dependence on Expensive Imaging Data:** While scan images are efficient, they are costly both financially and time-wise, limiting access for individuals with fewer resources.

- 7. Feature Overlap Problems:** Merging datasets may lead to feature imbalances, negatively affecting model learning, accuracy and precision.

By studying these gaps we look forward to introducing a more optimized and efficient deep learning algorithm with the help of computer vision that would support the clinical analysis by an early detection of Dementia in PD patients so that necessary steps are taken for the cure.

1.3 Problem Statement

Parkinson’s Disease seems to be transformed to Dementia in a slow and intriguing process which is less likely to be detected until it has reached a stage when the damage is too deep. Such delay in the diagnostic system plays a very vital role in damaging the patient’s condition from where the scope of improvement gets very little. Traditional systems of diagnosing like the MS-UPDRS for PD patients and tests like the Alzheimer’s Disease Assessment Scale (ADAS) or the MMSE used for Dementia depend immensely on subjective analysis. Not only are they time consuming and inconsistent, they seem to be delaying the process or giving inaccurate results.

In recent years, Artificial Intelligence has offered promising tools for disease tracking. For Parkinson’s, models using techniques like RNNs and LSTMs analyze time-series data such as gait patterns, handwriting, and sensor data. Similarly, for Dementia, CNN-based models process imaging data like PET, CT, and MRI scans to detect the alterations in brain structure and changes in its function. However, these tools typically address each condition independently, failing to capture their interconnected progression and shared mechanisms. Biomarkers such as hippocampal changes for Dementia and substantia nigra dysfunction for Parkinson’s are analyzed in isolation, which limits their potential for understanding how these diseases overlap and evolve together.

The challenge is further compounded by the lack of comprehensive datasets and integrated systems that consider the dynamic relationship between Parkinson’s disease, cognitive decline, and dementia. Many AI models also lack interpretability, making their outputs difficult for clinicians to trust and use effectively. Moreover, the reliance on advanced imaging technologies makes these solutions costly and impractical, particularly in resource-limited settings.

To address these issues, this study aims to develop an explainable, cost-effective deep learning model capable of detecting Parkinson’s Disease and then monitoring multiple-visit patient data from multiple MRI modalities as well as clinical features to predict the presence and degree of cognitive decline and thereby the onset of dementia. By focusing on optimizing existing diagnostic tools and addressing the overlap between these conditions, as well as by incorporating explainability tools to enhance clinical interpretability, the research seeks to enhance the accuracy and

reliability of early detection while making it more practical and accessible for real-world clinical use.

1.4 Objective

The main target of this study is to develop a framework with the combination of deep learning and computer vision with the data retrieved from a vast image set of T1 and T2-weighted MRI scan reports as well as clinical assessment scores and features of the same subjects, so that the detection of dementia is done accurately in confirmed patients of Parkinson's disease. Though there are several issues that make the research challenging, we look forward to improving upon the existing diagnosing procedures for a better result that could be used in the real world environment.

Here are some of the goals we aim to achieve :

- **Improving Accuracy :** Implementing advanced deep learning, single-modality, and multimodal techniques that could detect the subtle patterns and differences of the biomarkers which play an important role in the detection of the early stage of dementia.
- **Improving Usability for Clinicians:** Develop an interpretable AI model that offers clear insights about the presence of cognitive decline in particular brain regions that transitions into dementia, making it easier for healthcare professionals to make informed decisions.
- **Focusing on Scalability:** Design a solution that can be easily implemented across various healthcare environments, regardless of technological or resource constraints.
- **Monitoring Progression Over Time:** Continuous observation of image sets and clinical features for multiple visits over the course of 2-5 years to monitor any small or drastic changes to predict the outcomes in more effective ways.
- **Introducing Real-World Applications:** Merging the used algorithms and frameworks with the traditional clinical workflows to improve the diagnosing process and provide important support to the healthcare system in their decision making procedure.

By achieving these targets, this research attempts to minimize the gap between the advanced technologies and their practical applications in the healthcare system. The main focus is to create such a tool that could be helpful in finding the errors the traditional system cannot find to make the overall system more efficient for eventually to increase the betterment of the patient's condition. Using them in the practical scenario would benefit the patients widely to find their problem and cure them at an early stage while there is still time.

1.5 Methodology in Brief

In the system that we propose, our work is divided into two distinct phases; the first phase focuses on an optimized prediction of PD so that Healthy Control (HC) subjects can be completely filtered out.

Phase 1 covers Multi-Model PD Detection. We use subjects from the PPMI dataset in the form of a small subset of the PPMI that was retrieved from Kaggle to run training and testing. This subset contains T1 MRI scans of both Parkinson’s disease diagnosed patients (PD) and healthy controls (HC) in NIfTI format. The scans undergo preprocessing through GPU-accelerated resizing ($116 \times 116 \times 72$) and intensity normalization to $[0,1]$ range. The dataset is systematically organized with binary labels (HC=0, PD=1) and partitioned using a 70/15/15 split for training, validation, and testing with data augmentation applied to enhance model generalization. We then use five state-of-the-art deep learning architectures—ResNet-18, DenseNet-121, VGG-16, EfficientNet-B0, and Inception-V3—and train them, tuning hyperparameters in a manner that we can compare between their performances and identify the most effective baseline model for PD detection.

In Phase 2, we utilize different modalities of data from 175 filtered PD-only patients from the Kaggle subset, including T1-weighted MRIs, T2-weighted MRIs, and clinical features. Raw clinical data are processed and organized, and based on particular features such as NP1COG (which signifies that a higher score = worse cognitive decline) and MOCA_TOTAL (which signifies that a lower score = worse cognitive decline), cognitive decline is monitored across visits to assess its sustained manifestation and whether it worsens over time. This monitoring is formulated as a binary classification task, where a cognitive decline label of 1 indicates a sustained poor cognitive score for at least two consecutive visits for these features.

Carefully selecting the best models based on our Phase 1 analyses, which ended up being Inception-V3, EfficientNet-B0, and Resnet-18, we at first set a baseline by only training the model on T1-weighted scans using two-stage transfer learning by freezing all convolutional layers of the CNN architectures except the final two, to ensure controlled feature-extraction and fine-tuning when finding structural brain abnormalities. Then, we train different configurations of the modalities in the same way – T2-only, T1+T2. We incorporate multi-modality in our work by merging the T1-weighted structural MRI scan data with the clinical features dataset based on assessment scores of various clinical tests over multiple visits by the same 175 PD patients. Eight clinical features are selected, from which six derived features were computed, creating 48 derived trajectory features for each patient. Various methods of fusion, such as XGBoost, Bilinear Fusion, Concatenation and Attention Fusion are done. Models are trained over seven different configurations, from a full T1+T2+Clinical configuration and then systematically ablating each component to see which modality yields the absolute best results in correctly and robustly signaling cognitive decline, thereby pointing towards a future dementia diagnosis in PD patients and enhancing clinical decision-making. Attention-augmented explainability is built into the architecture; it generates attention maps that highlight the areas of the brain which have the most impact in predicting decline. This improves the

clinical interpretability and accessibility of the model results.

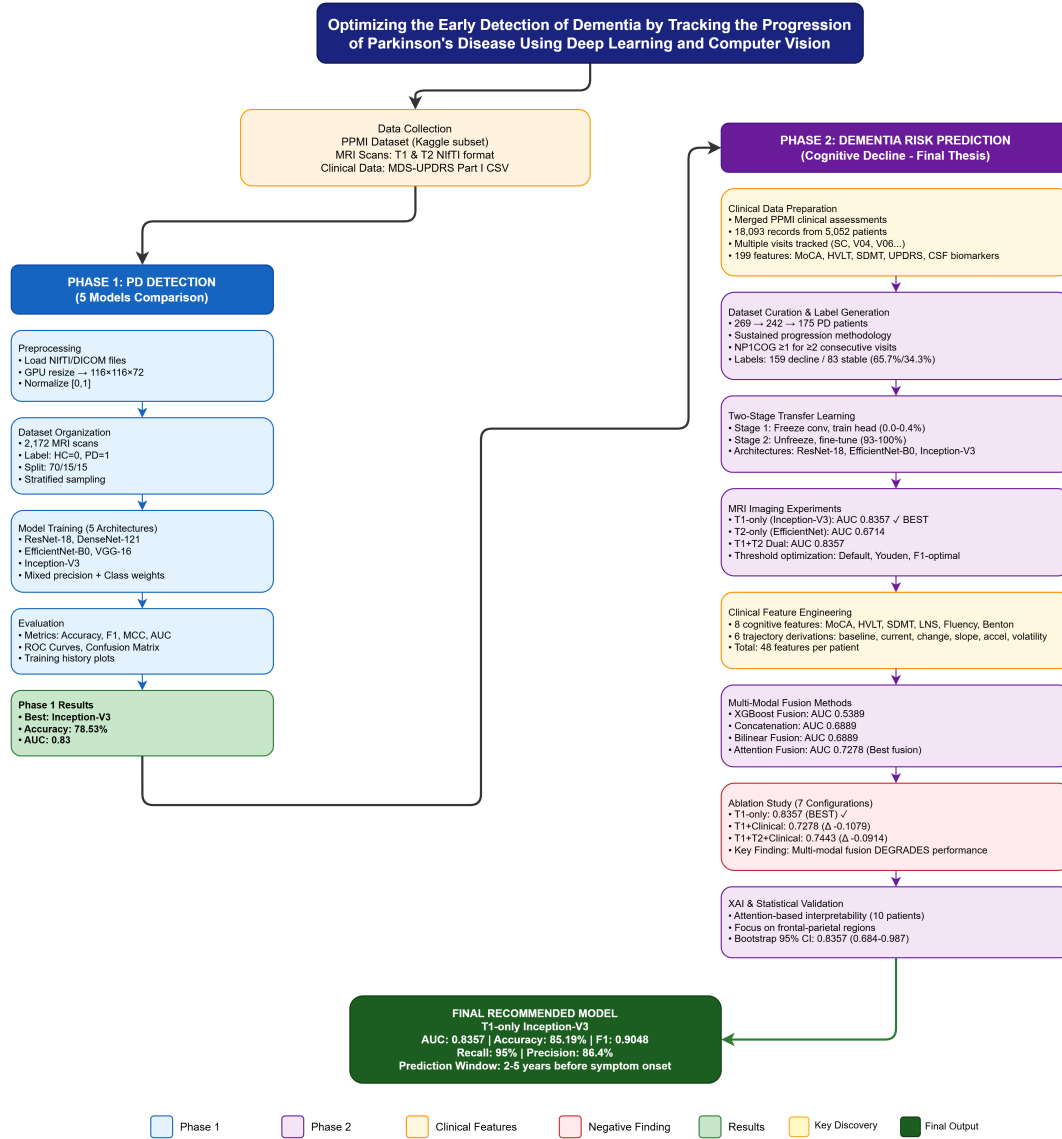


Figure 1.1: Workflow

1.6 Scopes and Challenges

Our research will aim to provide opportunities in the medical field and will try to maximize the diagnosis and treatment facilities for the betterment and the healthy living of mankind. Despite some wonderful prospects, there might be some challenges while our study is being processed which we will try to minimize.

• Scopes:

1. **Early Diagnosis of Dementia:** The early diagnosis of Dementia using an AI oriented system will enable physicians to detect the disease sooner which might play a crucial role in hindering the disease progression thus protecting the patients from cognitive impairment.
2. **Feature Extraction for PD Progression Analysis:** Specific features or biomarkers that lead to Dementia in PD patients can be extracted from the first model by training the model with images of MRI, CT Scan, PET etc. These extracted features can be assembled into a tabular dataset that provides opportunities for further research.
3. **Multimodal dataset:** The merged T1-weighted MRI dataset along with the 48 trajectory-based derived features help in understanding the situation of specific patients.
4. **Real-Time Tracking of PD Progression using Temporal Features:** Incorporating multiple-visit patient data over the course of several years enables real time monitoring of the image scans which might assist in the early diagnosis of the disease by observing the volumetric changes in those image scans which provides vital insights regarding neurodegenerative diseases.
5. **Real-World Applications:** This model can be used as an effective diagnostic tool in the future which might enable real time tracking of the disease and can be available to the mass population as mobile health applications or cloud based AI platforms.
6. **Explainability & Clinical Interpretability:** By implementing explainability and interpretability through tools like our inherent Attention mechanism, this will enable the physicians to understand the severity of the disease which might be crucial in providing precise and timely treatment to the patients.

• **Challenges:**

1. **Data Availability & Quality:** It will be a huge challenge to obtain datasets since there is a scarcity of datasets that contains features correlated to both PD and Dementia. Besides, acquiring quality image scans will be another challenge since we need to extract the features accurately to obtain precise results.
2. **Complexity of Disease Progression:** Tracking the progression of PD and leading to Dementia can be a complex process since the progression of these diseases can act in a non-linear manner. Sometimes, the health of the patients can deteriorate by suffering extreme cognitive impairment in spite of negligible changes in significant biomarkers.
3. **Computational Complexity:** Extracting features from raw image scans of MRI modalities can be computationally difficult. Besides, this process of extracting raw scans can be expensive since there might be a need for specific hardware resources and high processing power.
4. **Model Generalization & Bias:** Since most of the existing datasets that consist of correlated features of both PD and Dementia contain very

small samples and are also highly unavailable, the deep learning models might have a tendency to produce biased or non-generalized results due to the overfitting of the models.

5. **Handling False Positives and False Negatives:** False positives of Dementia prediction can lead to unnecessary stress or anxiety in patients that might lead to other fatal diseases. On the other hand, false negatives will disable the opportunities for the early diagnosis of the disease delaying the patient to take early treatment which might contribute to cognitive declination and worsening of the Dementia condition.
6. **Validation & Clinical Acceptance:** There might be significant challenges in gaining the trust of the physicians as the physicians might prefer the clinical diagnosis over the results provided by our cascading model. Our AI model has to prove it's reliability by providing validation against existing clinical methods.
7. **Ethical & Legal Considerations:** There might be a challenge in obtaining relevant image datasets since most hospitals don't provide the medical data of patients without the consent of the patients. Besides, we have to make sure that AI doesn't replace the human decision making system regarding the complex diagnosis of these diseases.

Chapter 2

Literature Review

2.1 Preliminaries

According to WHO, the neurological disorder that causes problems with movement, mental health, sleep, pain and other health issues is defined as Parkinson disease(PD). According to another study by Kouli et al. (2018), Parkinson's disease is a progressive neurodegenerative condition that not only affects movement during the initial stages, but also comprises a range of non-motor symptoms [26]. PD causes motor symptoms including slow movement, quivering, stiffness, difficulty in walking and losing balance. And the non-motor symptoms of PD are cognitive impairment, mental health and pain, sleep disorders, and sensory imbalances. Both neurologists and trained non-specialist healthcare personnel can diagnose Parkinson's disease(PD). The WHO reports that Parkinson's disease cannot be cured but its symptoms can be reduced with medication, surgery and rehabilitation. Parkinson's disease and other degenerative neurologic diseases can benefit from rehabilitation, especially physical therapy. Strength training, balance and gait training and hydrotherapy are among the treatments. These treatments can help individuals with Parkinson's disease function better and live better lives. This can also minimize the burden on their caregivers [2].

According to WHO, several diseases that affect memory, thinking, and the ability to perform daily activities fall under the term "Dementia". Common symptoms or signs of dementia include forgetting regular stuff or recent events, getting lost in random thoughts, being confused even in known environments, difficulties in decision making and difficulties performing daily activities etc. The patients with dementia mostly seem anxious, face personality disorder and deviate themselves from work or social activities. WHO further mentions that dementia is caused by a variety of diseases or lesions to the brain, with Alzheimer's disease being the most common among them. Other forms include vascular dementia, Lewy body dementia, and frontotemporal dementia. It can also result after a stroke, HIV infection, alcohol abuse, chronic traumatic encephalopathy or dietary inadequacies. The distinctions between various forms are blurred, and mixed forms frequently coexist. Dementia is a chronic illness that has no cure but it can be treated with physical activity, social connections and medicine [3].

Since we have studied the development of Dementia in PD patients in our paper, we have to be familiar with the terms Lewy Body Dementia (LBD) and Parkinson’s Dementia (PDD). Lewy bodies are aberrant protein aggregates that grow within nerve cells in the brain. The spread of Lewy bodies from subcortical to cortical areas has been significantly associated with dementia in Parkinson’s disease. Misfolded alpha-synuclein proteins generate Lewy bodies, which accumulate in neurons, particularly those in the cerebral cortex. This impairs normal neural function and contributes to cognitive impairment. PDD arises when cognitive functions are impaired. According to an existing study by (Huang, 2024), The clinically confirmed cases of dementia with Lewy bodies and Parkinson’s disease dementia are referred to as Lewy body dementia. It is characterized by chronic cognitive decline marked by cellular aggregates known as Lewy bodies which are located in the cytoplasm of cortical neurons. Parkinson’s disease dementia is the decline of cognitive abilities caused by Lewy bodies in the substantia nigra that arises in the later stage of the disease [27]. The important distinction is when certain symptoms first arise. Dementia symptoms appear within a year of movement symptoms in those with Lewy bodies. Parkinson’s dementia symptoms will occur after the normal Parkinson’s disease movements have been present for at least a year [5].

Deep Learning (DL) is a more advanced branch of machine learning that uses neural networks with multiple layers to automatically extract complex features from large datasets. DL excels in tasks like image recognition, natural language processing and medical diagnostics, often outperforming traditional ML methods in handling unstructured data such as images and text. Similar to the human brain, a neural network is a deep learning model that uses layers of interconnected nodes to process and identify patterns in data. The input layer, one or more hidden layers like convolution or pooling and the output layer called the fully connected layer are where each neuron transforms data using mathematical operations. Existing neural network techniques are Convolution Neural Network (CNN) which is mostly used for image recognition, pattern recognition etc, Recurrent Neural Network (RNN) which is mainly used for time series data or Long Short term Memory (LSTM) etc. Furthermore, the use of CV with imaging data is notable for the identification and progression of various disorders. Clinical scans such as PET, CT and MRI are processed by models that combine CNN and image data. The resulting analyses aid in the detection of changes in the brain’s structure and volume (engorgement or shrinking) as well as activity in various brain regions. Many studies have been carried out to use computer vision and image processing methods to extract disease biomarkers. An existing study by (A Computer Vision Approach to Diagnose Parkinson Disease Using Brain CT Images, 2018) claims that raw imaging data can be used to identify biomarkers for Parkinson’s disease. First they used the raw image scans as input and then that image was converted to grayscale. The algorithm proceeds to preprocess the image after which anisotropic filtering is applied. Following that, the picture segmentation of the Parkinson-affected area of the brain is completed. A bounding box is used and finally the desired results are attained [28].

2.2 Review of Existing Research

In a prior study published in *Psychiatry Research:Neuroimaging* (Morales et al., 2012), a Bayesian network classifier was used to predict dementia using MRI data from 45 Parkinson’s disease patients. Four classification models were utilized in this article: Naive Bayes, Multivariate Filter-Based Naive Bayes, Filter Selective Naive Bayes, and Support Vector Machines (SVM). They have been used to assess their ability to distinguish between cognitively intact patients with Parkinson’s disease (PDCI), PDMCI (Parkinson’s Disease with minimal cognitive impairment), and PDD. For this objective, 45 MRI scans (16 PDCI, 15 PDMCI and 14 PDD) were obtained and post-processed using Freesurfer. There were 112 variables (sub-cortical structure volumes and cortical parcel thicknesses) obtained per participant. A multivariate filter-based naive Bayes model was discovered to be the best classifier with the greatest cross-validated sensitivity of 0.9233, specificity of 1 and accuracy of 0.9655 [25].

A recent study published in Sage Journals (Tran & Byeon, 2024) predicted dementia in Parkinson’s disease patients using a small tabular dataset and a hybrid model. This work introduces LightGBM-TabPFN, a unique hybrid model for predicting dementia conversion in Parkinson’s disease. In this investigation, we observed the combination of LightGBM’s ability to handle missing values and TabPFN’s ability to exploit tiny datasets. LightGBM-TabPFN surpasses seven other approaches by obtaining exceptional accuracy and interpretability using SHAP analysis. It outperformed models like Extra Trees, Random Forest, Bagging Classifier, AdaBoost, XGBoost, HyperTab and LightGBM. This study uses data from 242 PD patients across 17 factors. The LightGBM-TabPFN model considerably outperformed seven other models. It has an accuracy of 0.9592 and an AUC of 0.9737. The interpretable LightGBM-TabPFN with SHAP represents a substantial improvement in neurodegenerative disease predictive modeling by giving clinical experts with insights into model predictions and opening up new avenues for clinical application [24].

Another recent study published in *Frontiers* (McFall et al., 2023) revealed major multi-modal markers of incipient dementia in Parkinson’s Disease (PD) using machine learning. The Random Forest classifier was used in conjunction with SHAP (SHapley Additive Explanations) for interpretation in order to uncover predictors that distinguish PD patients who later developed dementia (PDID) from those who did not (PDND). The study recruited 48 Parkinson’s disease patients (mean age 71.6 years, 44% female) and assessed 38 multi-modal predictors spanning ten domains including motor, cognitive, molecular, demographic, imaging and lifestyle characteristics. The model performed exceptionally well with an AUC of 0.84 and a Matthews Correlation Coefficient of 0.76. Tree SHAP analysis found that the top 10 predictors of incipient dementia explained 62.5% of the model’s variance. Motor symptoms (e.g., weaker gait), cognitive performance (e.g., slower Trail A), metabolic changes, ventricular capacity, age and daily activities were all significant predictors. The work demonstrates the potential of integrating machine learning with explainable AI to forecast dementia risk in Parkinson’s disease [29].

Another article from MDPI (Byeon, 2020) conducted a study that compared the non-motor symptoms of Parkinson’s Disease (PD) and Alzheimer’s Disease (AD) with an emphasis on identifying cognitive profiles using machine learning. The study used Parkinson’s Dementia Clinical Epidemiology Data (PADEM) from the National Biobank of Korea to create a prediction model using the Random Forest method. The model’s performance was evaluated against multiple logistic regression and Classification and Regression Tree (CART). With an average age of 74.6 years, the study comprised 228 individuals. 51.8% of whom had been diagnosed with Alzheimer’s disease (AD) and 48.2% with Parkinson’s disease dementia (PDD). The prevalence of PD (2.2%) and AD (4.4%) in the subjects’ families was low and 73.2% of them had only completed middle school. In addition, the prevalence rates for hyperlipidemia was 8.3%, high blood pressure was 46.3% and diabetes was 21.9% [30].

Another existing study which we found about from an article on the website Nature, used data from Sweden’s Parkinson’s Disease Swedish PD Registry (PARKreg) to examine the increased risk of dementia in PD patients. (Åström et al., 2022). The study linked data from 1581 Parkinson’s disease patients in Sweden’s southern area (Scania) to national health registries and compared them to a control group that was age and gender-matched. Kaplan-Meier plots and Cox regression were used to examine the time until dementia onset. During the study period, 683 people were diagnosed with dementia and 642 had multiple diagnoses in the registries. Approximately 47% of these diagnoses were made in primary healthcare settings, 29% in inpatient/outpatient settings and 37% in memory clinics. When dementia was defined as the use of anti-dementia medications (N06D), the number of cases increased to 780 however 160 people were not formally diagnosed with dementia at the end of the study [31].

Di Palo and Parde’s 2015 neural network model [32] used a CNN-LSTM architecture to detect different kinds of dementia by first detecting Alzheimer’s disease. Their model combined targeted features with implicitly learned features best optimized for disease detection. The Dementia Bank dataset was used for their study, and they used linguistic conversational transcripts. Local transcripts were covered by the CNN part of the model while LSTM covered long-term dependencies and sequential data. They passed their processed and extracted features through the model to assess performance metrics like the F1 score, and found a score of 0.929, which ushered in a new state-of-the-art at the time of the paper’s publication.

Couronné [33] introduced a refined model to track the progression of Parkinson’s disease, by first recognizing the existing markers of disease progression and recognizing the limitations and biases of current models. Most PD diagnoses and subsequent tracking are centered around motor symptoms, so datasets and cohorts related to the disease largely investigate only these symptoms, leading to temporal and population biases, although a much lesser number of non-motor symptoms like dysautonomia, sleep, and cognitive behavioral changes are also included. Cognition biases are also present in existing studies, owing to the fact that dementia is considered an exclusion criteria in datasets like Parkinson’s Progression Markers Initiatives (PPMI) and (Drug Interaction with Genes in Parkinson’s Disease) DIG-PD. However, dementia

in patients is a common side effect of PD, making patients with PD who ultimately develop dementia – that is, the subject of our study – a potential PD subtype whose exclusion removes dataset heterogeneity. In his paper, he mentions the use of IRMs, fMRIs, DATScan imaging records and striatal binding ratios derived therefrom to measure the loss of dopamine neurons in the substantia nigra, and biological samples which find the concentration of α -synuclein. Coronné’s test included 404 idiopathic PD subjects from the PPMI dataset composed of both imaging and clinical data; he proposed a deep longitudinal generative variational autoencoder spatiotemporal model, whose architecture consists of decoders that adapt to multimodal data inputs that can separate changes in a patient’s profile that emerge due to PD progression, from changes that occur due to inter-patient phenotypic variabilities. The model’s novelty is that it compensates for missing values by assuming data is Missing Completely at Random (MCAR), which eliminates the need for the imputation methods adopted by supervised models - thus reducing implicit biases - and allows joint modeling which improves accuracy of disease stage prediction.

Parashos et al. [34] conducted two multicenter, randomized and double-blind trial studies to track early Parkinson’s progression and identify potential disease-modifying agents using just clinical scales and instruments. Two NINDS NET-PD datasets (FS-1 and FS-TOO) were used, comprising 413 subjects with early untreated PD whose diagnosis was 5 years old, and who showed at least two of the three fundamental motor symptoms of the disease (bradykinesia, tremor, rigidity). The Unified Parkinson’s Disease Rating Scale (UPDRS) Parts II and III, to assess activities of daily living (ADL) and motor functions respectively, to understand the rate of disability and motor functional impairments in the subjects of their study. The mRS, the Schwab and England Independence Scale, the Total Functional Capacity (TFC) scale, the 12-item Short Form Health Survey (SF-12), and the 39-item Parkinson’s Disease Questionnaire (PDQ-39) ADL subscale were also used to measure disability. A linear model and a multivariable Cox proportional hazards model were used to determine results. The participants of the study were then randomized in a 1:1:1 scale to receive either creatine or minocycline for FS-1, either coenzyme Q10 or GPI-1485 for FS-TOO, or placebo. After the administration of their medication, their scale scores were measured either every three months, at the end-point, or at twelve months. By indicating the need of subjects to begin symptomatic therapy (ST) earlier, the UPDRS Parts II and III and the Schwab and England Independence scales were found to be highly sensitive in delineating disease progression in the subjects. However, tracking the disease in this manner may give rise to missing values problems, as well as a lack of generalizability.

Ribba et al. [11] used various computational and statistical models to build a simulation tool to track the progression of PD and predict the disease trajectory depending on different data-driven objective scenarios. To develop the model, longitudinal data was collected from 466 subjects from both PPMI and prodromal cohorts, excluding those carrying genetic forms of PD. Time-course data from MDS-UPDRS Parts I (non-motor ADL), II (motor ADL) and III (functional impairments) was used. PASADENA, a phase-2 randomized, double-blind placebo-controlled trial, was used to evaluate and validate the PPMI model predictions of the UPDRS data. The model’s input was the levodopa equivalent daily dose (LEDD) to incorporate the

effect of ST so that the data could be balanced as it tracked progression. A logistic progression model was used to compute the effect of ST on the calculated clinical score. It was found that the progression of UPDRS-Part III was thrice as fast as that of Part-I and II. The effect of disease-modifying treatment (DMT) on Part-III would precede that on Part-II by two to three years, and this could be utilized to study and innovate new DMT approaches or trials catered to each individual patient.

Kumar et al. [35] developed a dementia detection model combining machine learning and deep learning that uses speech feature analysis. They used publicly available speech data from the Pitt Corpus of the Dementia Bank database, a total of 442 audio recordings among which 218 were dementia patients and 224 were healthy control (HC) participants. Spectral noise gating was used to reduce noise, and normalization and segmentation were also done to preprocess the data, and then features were extracted like pitch, intensity, speech rate, pause duration, and more. After selecting the most relevant features and performing five-fold stratified cross validation, they were passed through ML classifiers like Support Vector Machines (SVM), k-Nearest Neighbors (KNN), Reduced Error Pruning Tree (REP), and Random Forests (RF) using the Weka toolkit. They were also passed through DL models like Artificial Neural Networks (ANN), CNNs, RNNs, and Parallel Recurrent Convolutional Neural Network (PRCNN) to capture complex speech patterns that may allude to dementia. Then performance metrics like accuracy, precision, recall and F1 score were calculated to evaluate the model’s capability of predicting dementia, with the highest importance being given to recall score because it determines the number of subjects who actually have dementia. The ML model performance accuracy ranges from 77.4% to 87.6%, while the DL model performances are more stable across architectures, ranging from 81.5 to 85%. Generalization could be more accurate if there was a much larger sample of speech data, but the ANN model gives better results for compact, simpler datasets. But overall, the best results in recognizing dementia from the Pitt Corpus dataset are from ML models (87.6%). Incorporating other data type features into the dataset in a multimodal manner would improve the scope of the detection model in this paper.

In a recent study [22], machine learning models and Deep Learning techniques like Feed-forward Neural Network(FNN), and K-nearest neighbors(KNN) models were used to identify the individuals with PD from a pool of healthy candidates based on voice signals analyzed from 195 voice recordings, collected from 31 different patients. Their model outperformed existing models at the time with a success rate of 99.11%. The study also mentions the current clinical landscape for PD, delineated by the National Collaborating Center for Chronic Conditions in 2006, highlighting the over-reliance on empirical testing, invasive techniques, and thorough clinical examinations [36]. These methods, unfortunately, pose challenges such as lack of efficiency, maintaining the costs, and limited potential for advancement.

Al-Fatlawi et al. [23] showed the diagnosis of PD using a Deep belief network (DBN) by testing and training voices collected from a number of healthy people and patients. In order to optimize the parameters, two stages of learning were applied: Unsupervised learning with Restricted Boltzmann Machines (RBMs) being the first stage to overcome the randomness of the initial weight values, while in the second

stage, supervised learning was used for fine tuning. With a 94% accuracy, this method - when being compared to other data-mining approaches like KNN, Ada-Boost, Random Forest, Support Vector Model (SVM) - proves to be an effective way to diagnose PD using speech signal data. The current clinical approaches for the detection of PD rely on some non-definitive tests based on symptoms like verbal glibness, issues in speech, or medical history of the patients - all of which are unable to detect PD in its early stages. Thus, there is a need for improved early detection methodology.

Li et al. [6] published an extensive paper reviewing 177 articles written between 2016 and 2021 that gave an idea about clinical approaches and a preview of AI's potential in the field of early dementia detection. The most prominent cause of dementia as per their studies is Alzheimer's disease (AD). The pathological alterations within the brain remain minimal in the preclinical stages of AD, showing signs of only mild cognitive impairment, but identifying each and every stage of these changes is imperative to improve the probability of success in drug trials and risk reduction strategies. According to their paper, the most commonly used clinical cognitive tests presently are the Mini-Mental State Examination (MMSE), the Addenbrooke's Cognitive Examination-Revised (ACE-R), the Montreal Cognitive Assessment (MoCA), and the Clock Drawing test (CDT). These tests measure the non-motor abilities of subjects by putting to task different cognitive functions, such as memory, fluency, attention, visuospatial coordination, etc. The paper finds that the MMSE shows 0.80-0.90 specificity while ACE-R, MoCA, and CDT show 0.84/0.89, 1.0/0.87, 0.85/0.85 sensitivity/specificity [37][38][39][40]. But a massive limitation of these tests, apart from CDT, is that a trained clinician has to be present to conduct them and determine their results [12]. The time taken to complete these tests is another downside. Moreover, score-based assessment tests overlook a notable amount of clinical data that might be related to cognitive processes. This is more prominent in the case of the clock drawing test. The quality of the drawing is not considered appropriately, thus, missing significant factors such as pausing, slowing down, and hesitations. So, the outcome becomes somewhat subjective [12]. However, AI offers promising results by utilizing biomarkers and other image data to develop a non-invasive and cost-effective early detection method of dementia. Some of the techniques are: 1. Computerized Cognitive Tests, 2. Movement Tests, 3. Speech, Conversation, and Language Tests, 4. Computer-assisted Interpolation of Brain Scans. To summarize, AI can help in the development of novel digital biomarkers for dementia detection in its early stages. Its ability to retrieve more features from a single test, reduce subjective judgment errors, and the ability to bring the automation of dementia screening to new heights along with quick evaluations of the cognitive functions of a large population can have a significant impact in this sector.

Hely et al. [4] followed up patients who were recently diagnosed with Parkinson's disease over the course of 20 years. Several methodologies were followed such as:

1. Diagnosing PD among de novo PD patients, who were on stages 1 to 3 on the Hoehn-Yahr scale, based on the presence of bradykinesia, resting tremor, and rigidity.
2. Examining in clinics and at home if required due to the increasingly frail body.

3. Extent of depression assessment based on the Geriatric Depression Scale (GDS).
4. Routine check to detect hallucinations and Illusions, amount of daytime sleepiness, and tests for cognitive functions such as MMSE and clock face drawings.
5. Interviews with family members to assess memory and cognitive abilities.

The results show that about 30 patients survived during this 20-year follow-up. The median survival was 54 months once dementia was diagnosed. During the 20th year, 83% of the surviving patients had dementia and 2 of the patients developed it afterwards. Moreover, 70 patients were found to have had the disease and died before the 20th year. For patients who had dementia at presentation (DLB), the average time of dementia diagnosis was 71.5 years while for those who were diagnosed later, it was about 71.6 years. Further data shows that 6 people were diagnosed during their 50's, 27 during their 60's, and 56 during their 70's. This shows a strong relationship between dementia development and age [41][42]. It was also found that the most important factor linking dementia with PD, regardless of Alzheimer's pathology, were the limbic and cortical Lewy bodies - stained using alpha-synuclein. However, it was also stated that neocortical Lewy body involvement doesn't ensure the correlation between dementia and PD. In this case, a high number of parahippocampal Lewy bodies and temporal bodies might provide a better correlation. However, in PDD, the number one clinical cause of dysfunction potentially lies within cellular mechanisms that are still relatively unexplored [43].

2.3 Summary of Key Findings

As explained in a blog (Dementia's Relationship With Parkinson's, n.d.), there are mainly two types of dementia that happen during the progression of PD. These are Lewy Body Dementia (LBD) and Parkinson's Disease Dementia (PDD) [5]. A cross-sectional study published in *Psychiatry Research: Neuroimaging* (Morales et al., 2012) to investigate the neuroanatomical biomarkers of dementia in PD patients was reviewed. To predict dementia, MRI data from 45 Parkinson's patients was analyzed using a Bayesian network classifier as explained in the study. The classifiers revealed that the sizes of the cerebral white matter, lateral ventricle and hippocampus were most closely linked with dementia in Parkinson's disease. Their research suggests that the volumetric measurements especially the expansion of the lateral inferior ventricle enlargement along with the shrinking of the hippocampus and cerebral white matter are the most effective neuroanatomical biomarkers in terms of distinguishing and predicting dementia in Parkinson's disease [25]. Even though the majority of the existing studies have pointed out the occurrence of both dementia types from Parkinson's disease, there hasn't been much research on the occurrence of both LBD and PDD together. According to a blog, both types of dementia may occur sequentially. Lewy bodies in Parkinson's disease first impact the brain region that controls movement, resulting in early symptoms such as difficulties initiating movement. As the disease continues, Lewy body damage spreads to regions of cognition and mental health causing memory loss, difficulty planning, concentration problems and mood swings. When these cognitive symptoms worsen, the disorder is called Parkinson's dementia [5]. According to another article published in *National Library of Medicine*, (Aarsland & Kurz, 2010), impairment in at least two of the four cognitive domains which are attention, non-cognitive processes, visuospatial decli-

nation and impairment of the recalling ability, that typically gets better with clues or suggestions is a common characteristic of cognitive deficiencies. The diagnosis of PDD is supported by the presence of at least one behavioral symptom such as excessive drowsiness during the day, being delusional, having visual hallucinations (VH), lack of interest in activities or a depressed or anxious mood. However, the diagnosis is still valid even if behavioral symptoms are absent [44]. According to various population-based epidemiological studies, 20 to 40% of people with Parkinson’s disease (PD) develop dementia. Incidence rates are 4-6 times higher than in non-PD controls. A recent Swedish study found that the 10-year cumulative incidence was roughly 54%. Other studies have found that at least 75% of persons with Parkinson’s disease who survive for more than ten years develop dementia [31]. Multiple articles have mentioned that people with PD are six times more likely to develop dementia compared to a healthy elderly [25][30][31][45]. According to another study, the development of dementia rises up to 83% in 20 years follow-up period [46].

The clinical examinations of Parkinson’s disease are conducted conventionally or manually using the Unified Parkinson’s Disease Rating Scale (UPDRS) and the Hoehn and Yahr scale. The most frequent tools for detecting dementia are standardized clinical tests that examine cognitive and executive functioning, memory, attention, distraction and so on as well as calculating minor cognitive impairment in early patients such as the Mini-Mental State Examination (MMSE). A study from Neurology uses the MMSE and the Hoehn and Yahr scales. According to Aarsland et al. (2001), Hoehn Yahr score >2 (OR, 3.4; 95% CI, 1.3 to 8.6) and Mini-Mental State Examination score <29 (OR, 3.3; 95% CI, 1.3 to 8.2) were predictive markers for dementia in Parkinson’s disease in addition to age at baseline [45]. One issue that has been identified in previous studies connecting Parkinson’s disease and dementia is a lack of adequate datasets as datasets having traits that correlate with both PD and dementia are not generally available. Furthermore, due to inherent biases in today’s models as well as the use of homogeneous and small datasets, the conclusions gained are not always relevant to broad general populations. To address the problem of small datasets, a variety of data augmentation approaches are applied. The study by Arnab et al. [47] demonstrates the use of data augmentation. This study used image augmentation to expand the amount of photos in a dataset. A dataset of 67 photos is used to produce 1000 images which improves the model’s accuracy. Augmentation offers new variants, reduces overfitting and increases accuracy on previously unknown data. It also stands up to occlusions, noise, scaling and orientation changes, boosting data efficiency and ensuring consistency across pictures and masks throughout transformation [48]. Along with surveys, interviews, case studies, clinical data, and other medical assessments using various scales to forecast Parkinson’s and dementia, various AI approaches, particularly ML, DL models and CV as well as image processing techniques are utilized to track disease development or discovery. However, the majority of these works rely on the independent diagnosis of diseases utilizing the aforementioned AI techniques. Previously, we saw the use of machine learning models to detect Parkinson’s disease and dementia. However, deep learning algorithms yield significantly more precise findings. We’ve also seen several studies using deep learning algorithms to detect Parkinson’s disease and dementia. Time-series data from gait analysis, wearable sensor data, handwriting and pen-holding as well as the progression of various motor and non-motor symptoms

are analyzed using neural networks models equipped with CNN, RNN, GRU, and LSTM techniques to determine the occurrence of PD to dementia. Furthermore, multiple pre-trained models such as CNN, ResNet, DenseNet and VGGNet are used to investigate volumetric changes in the cortical and subcortical regions of the brain. According to recent studies by Gao et al. (2019) and San-Segundo et al. (2019), various approaches have recently used convolutional neural networks (CNN) to represent parkinsonian patterns. CNN learns features at various levels of complexity, helping it to recognize more complex architectural patterns. In conjunction with gait analysis, a collection of wearable sensor signals such as vertical ground response force devices and accelerometers were employed to portray Parkinson’s disease in terms of force and acceleration connections [49][50]. Furthermore, time series-based CNN has been introduced for gait analysis in another study [51]. We also notice the presence of multimodal models for individual diseases. Efficient models have been created by incorporating several sorts of data, including gait analysis, handwriting skills, speech cognition, electronic signals, biomarkers and cognitive data. We have already seen multimodal predictors of dementia in Parkinson’s disease patients [29].

Chapter 3

Requirements, Impacts and Constraints

3.1 Final Specifications and Requirements

In this study, there were special technical and methodological criteria to create a deep learning system on the detection of Parkinson-like disease and the prediction of cognitive decline. The research was carried out in two periods. Phase 1 was dedicated to finding the best 3D convolutional neural network (CNN) architecture to use in PD detection, comparing the five possible models. Phase 2 took the most successful architecture as used in Phase 1 to predict the cognitive decline of the identified PD patients but with the help of Explainable AI (XAI) tools to improve clinical interpretability of the results.

Hardware Specifications: Training 3D CNNs was necessary to have high-performance computing resources to train on volumetric MRI data. A processor with 16 cores and 32 threads, AMD Ryzen 9 9950X was utilized in order to promote parallel processing of the data during preprocessing. Training on an NVIDIA RTX 5090 having 32 GB VRAM gave acceleration to model training and supported mixed-precision training with large batch sizes and high-resolution 3D images. The 64 GB DDR5 RAM and 1 TB SSD storage were included to support the system in terms of efficient access to memory and rapid loading of the data.

Software Requirements: PyTorch was used to implement the system with CUDA acceleration to train it in a more optimized way on a GPU. NiBabel was used in handling medical images (NIfTI files) and PyDICOM (DICOM formats). Numerical computation and data handling was done with NumPy and Pandas and preprocessing and evaluation measures with Scikit-learn. The XGBoost was utilized as tree-based feature fusion. In the case of visualization of XAI, Matplotlib and the custom attention- mapping modules were created to superimpose spatial weights of attention to MRI slices. The experimental setup was Python 3.8+, whereby package versions were carefully selected in order to make it reproducible.

Data Requirements: T1-weighted structural MRI image of both PD patients and healthy controls were used in Phase 1 in the PPMI data. Phase 2 involved diagnosed PD patients with 2-5 years longitudinal follow-ups of the patients and necessitated

both T1- and T2-weighted MRI scans. Images were all brought to 116 x 116 x 72 voxels. Phase 2 also involved the most detailed clinical aspects such as cognitive (MoCA, HVLIT, SDMT) and motor (MDS-UPDRS subscales, PIGD scores) and neuropsychiatric factors to facilitate multimodal fusion.

Methodological Requirements: The methodology focused on patient-level data partitioning in order to avoid information leakage, stratified sampling in order to maintain the balance in classes and systematic ablation studies in order to isolate the contribution of a single modality or architecture. The method of model evaluation was based on ROC-AUC analysis, confusion matrices, Matthews Correlation Coefficient (MCC) and optimized decision thresholds (Youden index and F1-optimal thresholds). Phase 2 also included attention-based XAI to visualize parts of the brain involved in prediction, which makes it clinically trustworthy and interpretable.

3.2 Societal Impact

Precise predictive models of the Parkinson disease detection and cognitive decline have significant implications on patients, families, and healthcare systems. PD is an illness with an estimated prevalence of 10 million cases across the globe with as many as 80% developing to suffer from Parkinson dementia (PD-D). Risky patients who may experience cognitive decline should therefore be identified early so as to be able to plan and intervene appropriately in the clinical setting.

Healthcare Benefits: The model of Phase 1 PD detection reached an accuracy of 78.53, which allows PD to be detected earlier with the help of structural MRI and possibly shorten diagnostic time. More importantly, the Phase 2 cognitive decline prediction model had solved a critical gap in the clinical literature because it was able to predict high-risk patients 2-5 years prior to the development of dementia. The model had great recall (95%), which puts the emphasis on early detection of those patients who are likely to suffer cognitive deterioration, and allows them to act proactively instead of reactively. The XAI that was based on attention offered visual images of contributing brain regions that were interpretable, enhancing clinician trust and clinical validation.

Patient and Family Impact: Cognitive decline risk information early enables patients and family to make informed decisions including financial, living arrangements, healthcare choices when patients have complete capacity to make decisions. Families are able to prepare both emotionally and practically as to the needs they may require in the future. XAI visualizations increase the creation of anxiety, as it provides clear descriptions of AI-generated predictions. Individual at-risk persons can be identified and then targeted in the clinical intervention of neuroprotective therapy. The fact that T1-weighted MRI alone has been found to perform well in predicting cognitive decline ($AUC = 0.8357$) indicates that expensive multimodal imaging might not be needed to predict cognitive decline. The consistency of attention maps showing frontal-parietal areas is also consistent with the known PD dementia pathophysiology, which supports the biological feasibility of the model.

3.3 Environmental Impact

The environmental impact of this research is minor but should be acknowledged. The combined training experiments of this study amount to 250-300 GPU hours on an NVIDIA RTX 4050 with 6 GB of VRAM. Assuming 140 W of power consumption per GPU, with a grid carbon intensity of 0.4 kg CO₂ per kWh, the carbon footprint is estimated at 14-17 kg CO₂-eq, comparable to driving 60-75 km in a car.

Some smart design decisions were made along the way. Early stopping ensured that the training process was not unnecessarily prolonged beyond the point of convergence, while caching the preprocessed data helped eliminate redundant computation, and sequential testing ensured that obviously suboptimal architectures were not trained. The efficiency of these designs, while greatly benefiting the research process, also had the secondary effect of reducing the environmental impact.

The major lesson learned from this research is that the simpler T1-only model performs better than the more complicated multi-modal approaches. If this model were to be deployed in clinical practice, this would mean that a one-modality system is sufficient, eliminating the need for additional T2 scans and possibly allowing for risk-based imaging.

3.4 Ethical Issues

The creation of predictive AI models and its implementation in healthcare also attract certain ethical concerns that underpin this study. The data were all taken by the Parkinson Progression markers initiative (PPMI) which is a publicly accessible dataset fully de-identified. Numerical codes (PATNO) substituted the identifiers of patients, and there was no effort to re-identify people. Data use agreement of PPMI was strictly followed in the study.

Algorithmic Fairness and Demographic Bias: In generated predictive reports demographic bias can be understood as consistent disparities in model functioning or confidence across groups of patients based on age, sex, ethnicity, level of disease, or socioeconomic standing. Since the model is a learned model based on historical cohort data, imbalance in the PPMI dataset can impact on patterns learned. Therefore, the reports produced by the model can be biased due to prevailing demographic patterns and not necessarily generalizable disease characteristics, whereas predictive stability and attention distributions can differ in underrepresented populations. Although the model can be effective at generating reliable and interpretable reports in well-represented groups, the same cannot be said about underrepresented populations. This bias is not based on individual bias at the population level. Reports must then be seen as speculative, cohort-based appraisals. The evaluation based on subgroups, metrics that are fair and external validation on a wide range of data should be included in the future works.

Clinical Responsibility and Explainability: These models are intended to assist and augment but not replace the decision-making process in medicine. In Phase 1 for Parkinson’s disease detection, the model achieved 78.53% accuracy, and in Phase 2 for cognitive decline prediction, it achieved an AUC of 0.8357. This indicates that the model performs well in terms of its predictive capabilities but is not perfect. Keeping in mind the need for transparency in healthcare AI systems from an ethical point of view, we have incorporated an XAI system based on attention that generates spatial attention maps for the MRI slices. This provides an overview of the brain regions that have the most impact on the model’s output for the clinicians to compare with their medical expertise and address the black box issue in medical AI systems. It is important for the clinical system to address the limitations by effectively communicating with the patient and monitoring them regardless of the output from the system. Patients should be made aware that the system is using AI for making predictions and what it entails, how the XAI system works in relation to the output generated by the model, and that they have the right to opt out from the AI-assisted assessment system.

3.5 Standards - if applicable

Established and accepted standards in medical imaging research, neuroimaging data management, and machine learning methodologies have been followed in this study. The MRI data abides by the Brain Imaging Data Structure (BIDS) naming standard, with subject-session-modality identifiers, for example, sub-3102_ses-BL_T1w.nii.gz. The MRI image scans are stored in the Neuroimaging Informatics Technology Initiative (NIfTI) data format, which is the standard data format in MRI research. Explainability is integrated into the model using the Attention mechanisms. Standard practices such as the use of a fixed random seed (seed=42) and deterministic CUDA operations (torch.backends.cudnn.deterministic=True) are followed to ensure reproducibility. Patient stratified splitting is also followed in this study to prevent data leakage and biased model evaluation.

3.6 Project Management Plan

This thesis has been carried out systematically by identifying certain tasks and dividing them into overlapping phases, which is required to accommodate the continuous nature of this work. The changes in the methodology design have been made numerous times to maintain compatibility with the available resources.

Project Schedule

The schedule delineating the entire project workflow over the course of a year has been represented in the form of the Gantt chart in figure 3.1.

Risk	Likelihood	Impact	Mitigation Strategy
Insufficient training data	Medium	High	Data augmentation, transfer learning from ImageNet
Class imbalance affecting training	High	Medium	Weighted loss functions, stratified sampling
Model overfitting on small dataset	High	High	Regularization, early stopping, attention mechanisms
Negative results from fusion	Medium	Medium	Prepared narrative for negative findings as valid contribution
XAI interpretations misaligned with neuroscience	Medium	High	Validation against known PD dementia pathophysiology
Reproducibility issues	Medium	Medium	Fixed seeds, deterministic operations, detailed logging

Table 3.1: Risk Management and Mitigation

Clinical features failed to enhance T1-only performance in Phase 2, confirming the anticipated risk of unfavorable outcomes from multi-modal fusion. This was handled by treating the result as a scientifically significant negative outcome, proving through rigorous methodology that on this dataset, simpler approaches perform better than complex fusion. Crucially, the XAI risk was effectively reduced: attention maps consistently emphasized frontal-parietal cortical regions and subcortical structures, confirming the biological plausibility of model decisions and matching established neuroscientific understanding of PD-related cognitive decline.

3.8 Economic Analysis

The proposed system demonstrates high cost-effectiveness; little to no costs were drawn up in the research implementation phase. If deployed clinically, potential savings of \$65,000-176,000 could be made per patient, as the models could generate:

1. automated PD screening reducing specialist consultation time;
2. estimated 20-30% reduction in routine MRI scheduling for low-risk patients who can be assured to not need follow-ups;
3. early identification before cognitive decline or dementia symptoms start showing, thus opening intervention windows for neuroprotection and preventing costly crisis care for cognitive emergencies;
4. more efficient specialist allocation based on risk stratification;
5. an automated analysis which reduces evaluation time from 60 minutes to just 10;

6. a 95% recall score which means the number of missed at-risk cases would be minimal;
7. an 86.4% precision score which means the reduction of false alarms, thereby the reduction of possible patient anxiety

Besides, the in-built Attention-based explainability component of the model adds value by reducing the need for additional diagnostic efforts when attention maps confirm clinically expected patterns, and by enabling faster clinical acceptance of AI-assisted diagnosis, as well as gaining the trust and validation of clinicians.

Cost Benefit Estimation

Developing and setting the system in place in clinical environments would incur reasonable costs stemming from regulatory preparation, software development and licensing, external validation study, IT security compliances, GPU and server hardware, staff training, workflow optimization, cloud computing, quality assurance, and technical support, but these would be one-time costs, routine maintenance notwithstanding.

After deployment, application of the entire system from start to finish is quick and effective, as demarcated in Table 3.2.

Step	Action	System	Time
1	PD patient undergoes routine T1-weighted MRI	MRI Scanner	8-12 min
2	DICOM images auto-routed to AI pipeline	PACS	< 1 min
3	Model generates risk score and attention map	AI Server	2-3 min
4	Results delivered to neurologist workstation	EMR/PACS	< 1 min
5	High-risk patients (>60%) flagged automatically	EMR Alert	Immediate
6	Neurologist reviews AI output with clinical context	Workstation	5-10 min
7	Care plan adjusted based on risk stratification	EMR	Variable
Total Turnaround	From scan completion to results	End-to-end	< 15 min

Table 3.2: Time-Saving System Application

The key economic finding from Phase 2 is that T1-only imaging provides optimal prediction, eliminating the need for additional T2 acquisitions in cognitive decline risk assessment, representing a potential 40-50% reduction in imaging costs compared to multi-modal protocols while achieving superior predictive accuracy. Rigorous ablation study demonstrating that simpler approaches outperform complex fusion provides important guidance for the field.

Advantage	Description	Benefit
No New Imaging Required	Uses existing T1 MRI protocols	Zero additional acquisition cost
PACS Integration	Fits existing radiology workflow	Minimal workflow disruption
Interpretable Outputs	Attention maps for clinical validation	Enables clinician trust and adoption
Scalable Architecture	Computational, not specialist-limited	Unlimited throughput capacity
Superior Performance	AUC 0.8357 vs 0.7443 multi-modal	Better predictions, simpler system
Cost-Effective	40-55% cheaper than alternatives	Lower barrier to adoption
Rapid ROI	>4,000% first-year return	Investment recovered in < 1 month

Table 3.3: Cost-Benefit Analysis and ROI

Chapter 4

Proposed Methodology

4.1 Design Process or Methodology Overview

The aim of this research is the early detection of cognitive decline that ultimately manifests as dementia in diagnosed Parkinson’s Disease (PD) patients following an experimental design with two connected deep-learning phases. We structure our complete methodology as such: 1) the accurate classification of PD from structural brain MRI to demonstrate the feasibility of automated neurological disease detection, which is the first step to a confirmed detection; and 2) the prediction of cognitive impairment in PD patients using multimodal structural brain MRI and clinical features and combined with Attention-enhanced explainability AI (xAI) for clinical interpretability and model validation.

The step-by-step process employs a subset of the Parkinson’s Progression Markers Initiative (PPMI) dataset obtained from Kaggle [52] to accomplish the task of PD detection. This dataset contains three-dimensional volumes of cranial T1 and T2-weighted MRI in NIfTI format of both PD diagnosed individuals and healthy controls. T1-weighted scans were utilized for Phase 1 PD Detection. Both T1 and T2-weighted scans were utilized for Phase 2 for cognitive impairment prediction, combined with temporal clinical assessment data scores based on the Movement Disorder Society United Parkinson’s Disease Rating Scale (MDS-UPDRS) Part I, directly obtained from the PPMI database [53].

Phase 1 - PD Detection: The objective was to classify structural MRI scans as either Parkinson’s Disease (PD) or Healthy Control (HC) using 3D convolutional neural networks. Five architectures were systematically evaluated: ResNet-18, DenseNet-121, EfficientNet-B0, VGG-16, and Inception-V3. All models were adapted from 2D to 3D convolutions and trained using ImageNet-pretrained weights where applicable. This phase established Inception-V3 as the best-performing architecture with 78.53% test accuracy and AUC-ROC of 0.8252, providing the architectural foundation for Phase 2.

Phase 2 - Cognitive Decline Prediction: In order to identify the most efficient way to predict future cognitive decline in PD patients, the T1-weighted MRIs of PD-only subjects were trained through two-stage transfer learning using three 3D convolutional models (Inception-V3, EfficientNetB0, and Resnet-18) that were se-

lected based on the experimental comparisons obtained from the model evaluations from Phase 1. The best-performing model on the T1 modality was Inception-V3, with an AUC of 0.8357, which was selected as the baseline for further comparison. Multiple imaging modalities were also tested, such as T2-only and T1+T2. Raw clinical assessment data of the same 175 subjects was processed, and cognitive decline was evaluated in a binary manner. Specific features like NP1COG (higher score = worse cognitive decline) and MOCA_TOTAL (lower score = worse cognitive decline) were given emphasis, and cognitive decline was observed over the course of 2-5 year visit data to see whether or not it got worse; cognitive decline label 1 was assigned to imply a sustained bad cognitive score for at least 2 consecutive visits for those features. Multi-modality was brought in by fusing clinical features with the T1-weighted MRI modality, using eight best features that were picked by feature selection: moca_total, hvlt_total_learning, fluency_total, sdmt_total, lns_total, hvlt_delayed_recall, benton_jlo_total, and fluency_animals. Four fusion methods were tried: XGBoost, Bilinear Fusion, Concatenation Fusion, and Attention Fusion, with Attention Fusion giving the best performance. A three-way configuration with T1+T2+Clinical was also implemented. A complete ablation study of seven different configurations showed that T1-only remained the best-performing modality, introducing the aspect in our research that single-modality testing gave the most discriminative outcomes for cognitive prediction.

The strong clinical connection between PD and dementia is the drive behind this two-phased approach: 40-80% [54] of PD patients develop dementia in later stages, making PD a prodromal marker for cognitive decline. Brain imaging biomarkers in PD correlate with dementia risk, so our process aims to first demonstrate robust PD detection capabilities and then use similar structural imaging modalities fused with clinical features to examine brain changes over multiple visits to identify cognitive impairment. Attention mechanism was built into the architecture to generate maps that display which brain regions contribute to cognitive decline prediction, and these maps make the final outputs interpretable by clinicians so that the results of our work are accessible to and helpful for more people.

The design process is constrained by computational resources, lack of complete desired dataset availability, and the need for clinically interpretable results. To address these constraints, the methodology employs the limited Kaggle subset of PPMI, GPU-optimized deep learning pipelines, automatic memory management with graceful fallback mechanisms, and comprehensive comparative evaluation across multiple state-of-the-art 3D convolutional neural network architectures.

4.2 Preliminary Design or Design (Model) Specification

4.2.1 Phase 1: Parkinson’s Disease Detection (Multi-Model Comparison)

The objective was to classify structural MRI scans as either Parkinson’s Disease (PD) or Healthy Control (HC) using 3D convolutional neural networks. In order to provide a comprehensive comparative baseline, five distinct 3D CNN architectures were selected that each represented different design philosophies in deep learning and are known to perform excellently in the field of medical image classification: ResNet-18, DenseNet-121, EfficientNet-B0, VGG-16, and Inception-V3. All models were adapted from 2D to 3D convolutions and trained using ImageNet-pretrained weights where applicable. The fundamental characters of the chosen architectures are mentioned below:

ResNet-18 (3D)

This model implements residual connections enabling training of deeper networks without degradation. Its architecture consists of four residual blocks with channel progression ($64 \rightarrow 128 \rightarrow 256 \rightarrow 512$). There are approximately 33.16 million parameters. The model utilizes skip connections that facilitate gradient flow and feature reuse. The vanishing gradient problem is addressed through identity mapping by this model while keeping up computational efficiency.

DenseNet-121 (3D)

The model uses dense connectivity where each layer receives input from all preceding layers within a block. These dense connections create implicit deep supervision and reduce the number of parameters, around 11.24 million, compared to traditional architectures. Its design maximizes information flow through the network and encourages feature reuse, with a growth rate of 32 and bottleneck layers for computational efficiency and increased feature hierarchy learning.

EfficientNet-B0 (3D)

This model utilizes mobile inverted bottleneck convolutions (MBConv blocks) with squeeze-and-excitation attention mechanisms for channel-wise feature recalibration. With a more compact design, its architecture achieves high accuracy with fewer parameters, approximately 4.07 million, through compound scaling of depth, width, and resolution, incorporating Swish activation functions for improved gradient flow.

VGG-16 (3D, Reduced)

This model is an adaptation of the classical VGG architecture with reduced channel counts (divided by 2) to manage 3D memory requirements. It is an extremely large model with a deep stack of $3 \times 3 \times 3$ convolutional filters with batch normalization that demonstrate the effectiveness of small receptive fields with increased depth.

Parameter count is significantly higher, around 15.66 million. Gradient checkpointing is used to reduce memory consumption during backpropagation, even so, it is a computationally intensive model that demands significant VRAM.

Inception-V3 (3D)

This model uses Inception modules and implements a 48-layer deep CNN with parallel convolutional filters of varying sizes ($1 \times 1 \times 1$, $3 \times 3 \times 3$, $5 \times 5 \times 5$ factorized) within each module, enabling multi-scale feature extraction. It has approximately 14.44 million parameters and is memory-optimized. Its architecture captures both local and global spatial patterns simultaneously, with auxiliary classifiers providing additional gradient signals during training, promoting diverse feature extraction.

All architectures above use Kaiming-Xavier initialization for weight initialization, batch normalization for training stability, adaptive global average pooling for spatial invariance, and ReLU/Swish activations for non-linearity. The final classification layer outputs 2 classes (Healthy Control vs. Parkinson’s Disease) with softmax activation.

This phase established Inception-V3 as the best-performing architecture with 78.53% test accuracy and AUC-ROC of 0.8252, providing the architectural foundation for Phase 2.

4.2.2 Cognitive Decline Prediction in existing PD Patients (Multi-Modal Fusion and Ablation Studies)

4.2.2.1 Architecture Selection for Phase 2

The comparative experimental study performed in Phase 1 for the preliminary task of PD detection from structural MRI informed the architecture selection for Phase 2 for cognitive decline prediction. Inception-V3 achieved the highest test accuracy (78.53%), F1-score (0.8589), MCC (0.4159), and AUC-ROC (0.8252), far outperforming all other models. The VGG-16 model had the lowest accuracy (61.35%) and AUC-ROC (0.6581), and was eliminated from consideration to prevent massive overfitting. ResNet-18, DenseNet-121, and EfficientNet-B0 were the middle-tier performers, their AUC values ranging from 0.69 to 0.74. Training dynamics revealed extreme variance, with accuracy fluctuating between 60 and 85% even in later epochs, along with low MCC scores (none exceeding 0.42). This indicated that the characteristics of structural MRI that differentiate PD brains from normal brains are very subtle, thus hard to learn. In the end, Inception-V3 was selected as the primary architecture for Phase 2 cognitive decline prediction due to its superior discriminative ability and multi-scale feature extraction capability. ResNet-18 was retained as a capacity baseline, and EfficientNet-B0 as the parameter-efficient alternative.

4.2.2.2 Systematic Evaluation of Transfer Learning

A two-stage transfer learning protocol was implemented, first on solely T1-weighted MRI scans of PD-only subjects to establish a baseline:

1. Stage 1 (Feature Extraction): All layers frozen, except the final two for rapid convergence (0.0-0.4% trainable parameters).
2. Stage 2 (Fine Tuning): Deeper layers progressively unfrozen for domain adaptation to MRI-specific features (93-99% trainable parameters).

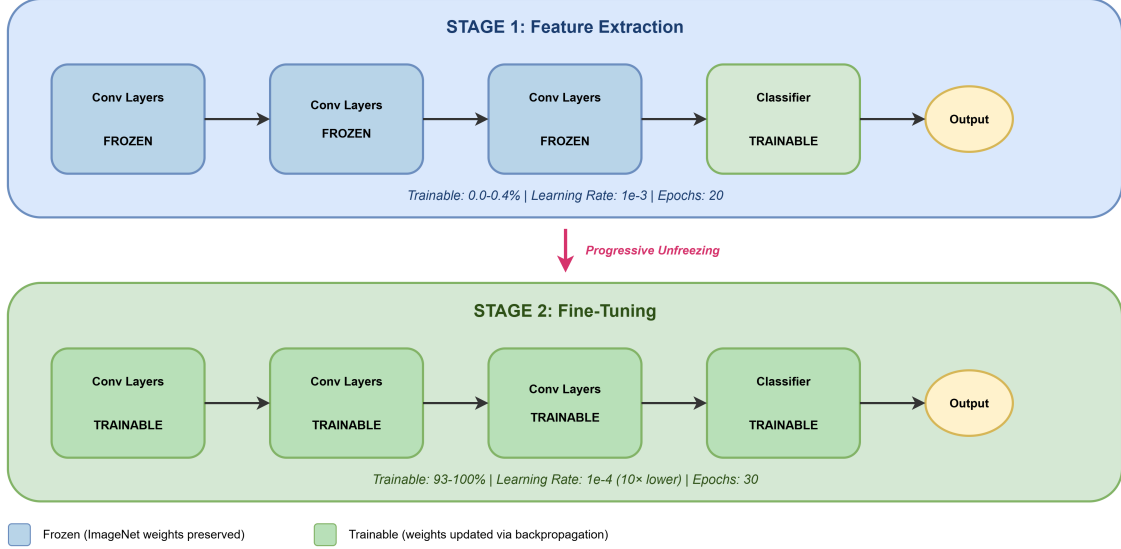


Figure 4.1: Two-Stage Transfer Learning

4.2.2.3 Spatial Attention Mechanism for Explainability

An Attention-augmented XAI was integrated into the architecture to generate interpretable attention maps, in order to provide spatial visualization of brain regions that contribute to predictions. This allows the results to translate directly to clinical practice, encouraging neuroscientific validation.

Attention Module Architecture

Given feature map $F \in R^{C \times D \times H \times W}$ from the final convolutional block:

$$F_{spatial} = AvgPool_{channel}(F) \in R^{1 \times D \times H \times W} \quad (4.1)$$

$$A = \sigma(Conv_{1 \times 1 \times 1}(F_{spatial})) \in [0, 1]^{1 \times D \times H \times W} \quad (4.2)$$

$$F' = F \odot broadcast(A) \quad (4.3)$$

where σ is the sigmoid activation and $\odot$ represents element-wise multiplication.

Table 4.1 represents the regions of the brain that typically highlight cognitive function.

Brain Region	Known Role in PD Cognition	Expected Attention
Prefrontal Cortex	Executive function, working memory	High (0.7-0.9)
Posterior Parietal Cortex	Visuospatial processing	High (0.6-0.8)
Hippocampus	Memory consolidation	Moderate-High (0.5-0.7)
Caudate Nucleus	Frontostriatal circuits	Moderate (0.4-0.6)
Temporal Cortex	Semantic memory	Moderate (0.4-0.6)

Table 4.1: Expected Attention Distribution Based on Literature

4.2.2.4 Multi-Modal Fusion of T1+Clinical Features

Multimodality was introduced into the architecture by integrating T1-weighted imaging features with clinical trajectory data. Eight best clinical features were selected, then four fusion strategies were designed and evaluated, considering theoretical suitability for our dataset constraints (n=175 patients, 122 training samples). The fusion strategies were:

1. XGBoost Fusion, which extracts CNN features from pretrained Inception-V3 and combines them with clinical features using gradient-boosted trees.
2. Concatenation Fusion, which implements late fusion by concatenating projected feature vectors.
3. Bilinear Fusion, which uses an nn.Bilinear layer to model pairwise interactions between modalities.
4. Attention Fusion, which learns adaptive weights to determine the relative importance of each modality.

Table 4.2 depicts a theoretical comparison of the four methods’ mechanisms and outputs, in accordance with their compatibility with this research:

Fusion Method	Parameters	Mechanism	Small Data Suitability	Interpretability
XGBoost	~500 trees	Gradient-boosted decision trees on concatenated features	Good - native missing data handling	High - feature importance
Concatenation	~10K	Simple vector concatenation followed by MLP classifier	Good - minimal parameters	Low - black box MLP
Bilinear	~262K	Pairwise multiplicative interactions via outer product	Poor - high overfitting risk	Medium - interaction matrix
Attention	~15K	Learned adaptive weights for each modality via softmax	Good - can suppress noise	High - attention weights

Table 4.2: Comparison of Multi-Modal Fusion Alternatives

The Bilinear fusion with 2,147 parameters per sample dramatically exceeds safe thresholds, predicting severe overfitting. The Attention Fusion approach was particularly important, as it provided dual interpretability: modality-level attention

showed the relative importance of imaging vs. clinical features, and spatial attention showed which brain regions drive the imaging component’s predictions.

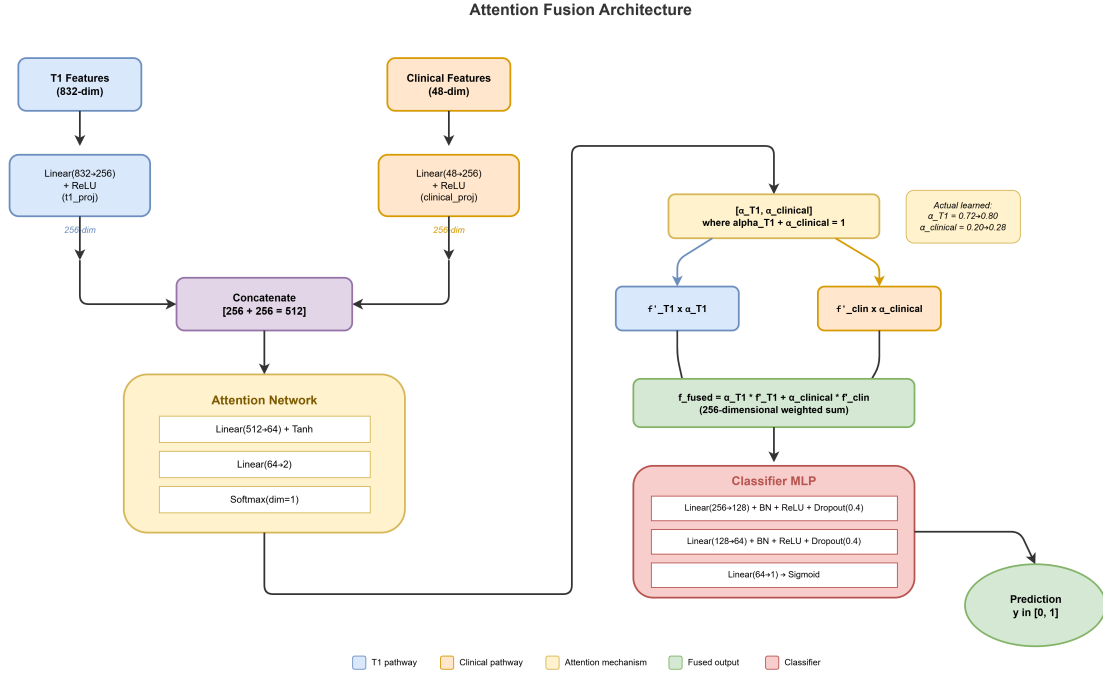


Figure 4.2: T1+Clinical Attention Fusion Architecture

Three-way Attention Fusion was also applied, wherein three modalities, T1+T2+Clinical Features, were fused for a full multi-model configuration. Every configuration of the three modalities in this research, and the architectures used to implement them, has been evidenced in Table 4.3:

Configuration	Architecture	Modalities	Parameters
T1-only	Inception-V3 + Spatial Attention	T1 MRI	23.9M
T2-only	EfficientNet-B0 + Spatial Attention	T2 MRI	5.4M
T1+T2 Dual	Dual Inception-V3 + Attention	T1 + T2 MRI	47.8M
Clinical-only	MLP (Linear→64→32→1)	18 features	~3K
T1+Clinical (XGBoost)	Inception-V3 + XGBoost	T1 + Clinical	23.9M + trees
T1+Clinical (Concat)	Inception-V3 + Concat MLP	T1 + Clinical	23.9M + ~10K
T1+Clinical (Bilinear)	Inception-V3 + nn.Bilinear	T1 + Clinical	23.9M + ~2.1M
T1+Clinical (Attention)	Inception-V3 + Attention	T1 + Clinical	23.9M + ~15K
T1+T2+Clinical	Dual + 3-way Attention	All modalities	47.8M + ~20K

Table 4.3: Complete Model Configuration Summary

Moreover, projection dimension was kept 256 for all modalities to maintain fair attention weighting, BCE loss function with class weights (1:92:1) was used to handle 65.7% decline vs 34.3% stable imbalance, and regularization techniques like keeping dropout rate=0.4 combined with batch normalization were used to prevent overfitting on n=175.

4.2.2.5 Ablation Studies and Fusion Method Comparison

Holding the T1-only AUC as the baseline, modalities were removed systematically one-by-one from the T1+T2+clinical full configuration to understand the impact of each modality on the predictions. Table 4.4 shows the seven configurations tested in the study along with the expected outcome.

Configuration	Purpose	Expected Outcome Based on Simulations
T1-only (Inception-V3)	Establish imaging baseline	Highest AUC expected (~ 0.84)
T2-only (EfficientNet-B0)	Evaluate T2 modality contribution	Lower AUC expected (~ 0.67)
T1+T2 Dual	Test imaging modality fusion	No improvement over T1-only expected (~ 0.84)
Clinical-only (MLP)	Evaluate clinical features alone	Moderate AUC expected (~ 0.75)
T1+Clinical (Attention)	Test imaging-clinical fusion	Degradation expected (~ 0.73)
T2+Clinical (Attention)	Test weaker imaging with clinical	Low AUC expected (~ 0.65)
T1+T2+Clinical (3-way)	Full multi-modal fusion	No improvement expected (~ 0.74)

Table 4.4: Ablation Study Results - Fusion Methods vs T1-only Baseline

4.3 Data Collection

Data was obtained from the Parkinson’s Progression Markers Initiative (PPMI), accessed through Kaggle (<https://www.kaggle.com/datasets/beterramzy/parkinsons-data/data>) and the official PPMI website. For Phase 1 (PD Detection), the dataset comprised T1-weighted MRI scans including both PD patients and healthy controls with balanced class distribution. For Phase 2 (Cognitive Decline Prediction), which represents the primary research objective addressing early dementia detection, 269 subjects were initially available, with T1-weighted and T2-weighted structural MRI scans.

For the clinical features, a comprehensive clinical dataset was constructed by merging multiple PPMI data files by integrating on a patient-level basis. PATNO, the dataset’s patient identifier, and EVENT.ID, the dataset’s visit identifier, were used as merge keys, and ten primary CSV files were combined thereafter: MDS-UPDRS Parts I-III for motor and non-motor assessments, Montreal Cognitive Assessment (MoCA) for global cognition, Hopkins Verbal Learning Test (HVLT-R) for memory, Symbol Digit Modalities Test (SDMT) for processing speed, Semantic Fluency tests for executive function, Geriatric Depression Scale (GDS-15), State-Trait Anxiety Inventory (STAI), and MRI scan metadata connecting imaging files to corresponding clinical visits. Baseline MRI scans were temporally aligned to the closest clinical assessment within a 90-day window. The earliest qualifying baseline visit was selected to maximize follow-up duration, with a mean of 2.7 years. In the end, the merged dataset contained 199 available clinical features.

4.3.1 Data Cleaning

Phase 1

MRI scans were validated for completeness and quality. Images with artifacts or incomplete brain coverage were excluded. The final Phase 1 dataset maintained

balanced representation of PD and HC classes for unbiased model training and evaluation.

Phase 2

The research study initially had 269 participants, which was reduced after a series of filtering processes. First, 27 participants were eliminated since they did not have enough longitudinal information, i.e., fewer than two clinical visits after a year. Then, 67 healthy controls were eliminated to ensure that the research focused on predicting cognitive decline in Parkinson’s disease only, i.e., which Parkinson’s patients would develop from normal cognition to cognitive decline. After these processes, the Phase 2 study population consisted of 175 diagnosed PD patients with enough follow-up time (average of about 2.7 years, ranging from 1.2 to 5.8 years).

Handling of Missing Data

First, we examined the clinical features for missing information. CSF biomarkers, i.e., A42, -synuclein, p-tau181, and total tau, were mostly missing (59-77%) since the lumbar puncture is a highly invasive procedure. We did not use this information in the study but kept it optional for secondary studies. Motor scores, i.e., MDS-UPDRS Parts I-III, had 8-12% missing values, mostly because of missed visits. Cognitive tests showed the following percentage of missing values: MoCA 3%, HVLT-R 8%, SDMT 11%, Letter-Number Sequencing 15%, and Benton Judgment of Line Orientation 17%. We used appropriate imputation methods for the cognitive tests. Patients with >20% overall missingness across selected features (n=8) were retained but flagged for sensitivity analysis to assess their influence on model performance.

Handling of Outliers

We reviewed the clinical scores against the known score ranges for the instrument (e.g., MoCA: 0-30, MDS-UPDRS Part III: 0-132, GDS-15: 0-15). We reviewed values that could not possibly fall within the physiological limits of the instrument against the next available visit. We corrected any obviously erroneous values (e.g., a patient with a MoCA score of 47).

4.3.2 Data Transformation

Image Standardization and Resizing:

All the MRI volumes were geometrically and intensity normalized to make them consistent and comparable to each other. The geometric transformation dealt with the non-homogeneous spatial dimensions of the unprocessed NIFTI files, which ranged between 176x 208x 176 voxels and 256x 256x 166 voxels (voxel counts) in accordance with the protocol used in the acquisition procedure. All the volumes were downsampled to a uniform size of 116x116x72 voxels, which are the standard size of

the input to the 3D CNN models. Such size was chosen to have a balance between maintaining enough spatial resolution particularly in cortical and subcortical areas of interest to the pathology of Parkinson disease and compatibility with network designs that stack layers in powers of two. Normalization of intensity has been done to ensure that inter-scan variations due to differences in MRI scanners and acquisition parameters are countered. The 1st to 99 th percentile values were chunked out to remove outliers, and then the values were scaled to the $[0, 1]$ range through a linear transformation. This method of normalization was robust and guaranteed similar intensity distributions in all scans and relative tissue contrast at the same time. Below, we have provided visualizations of 10 sample images before and after resizing.

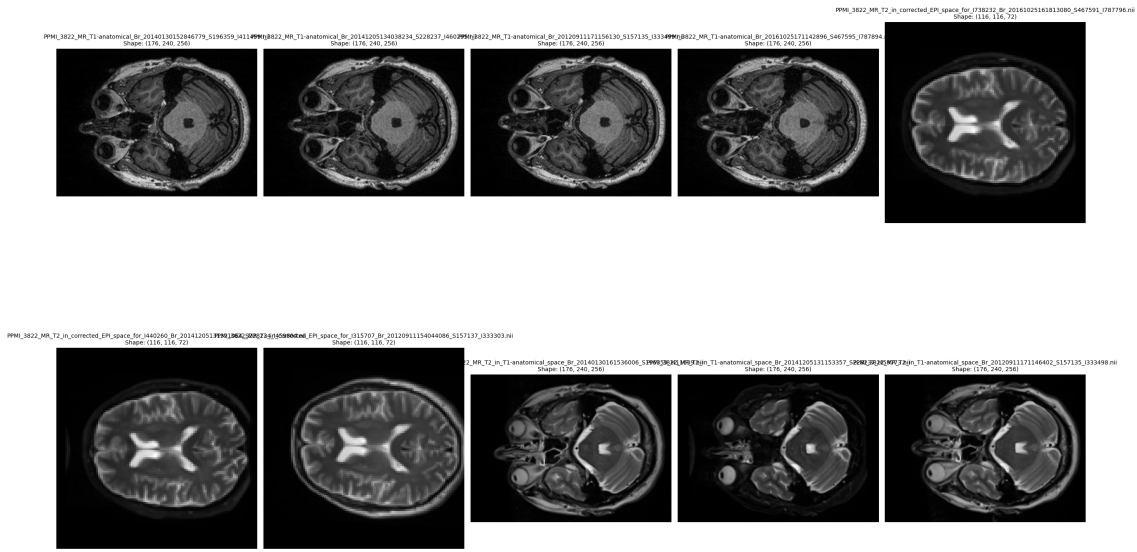


Figure 4.3: MRI Scans Before Resizing

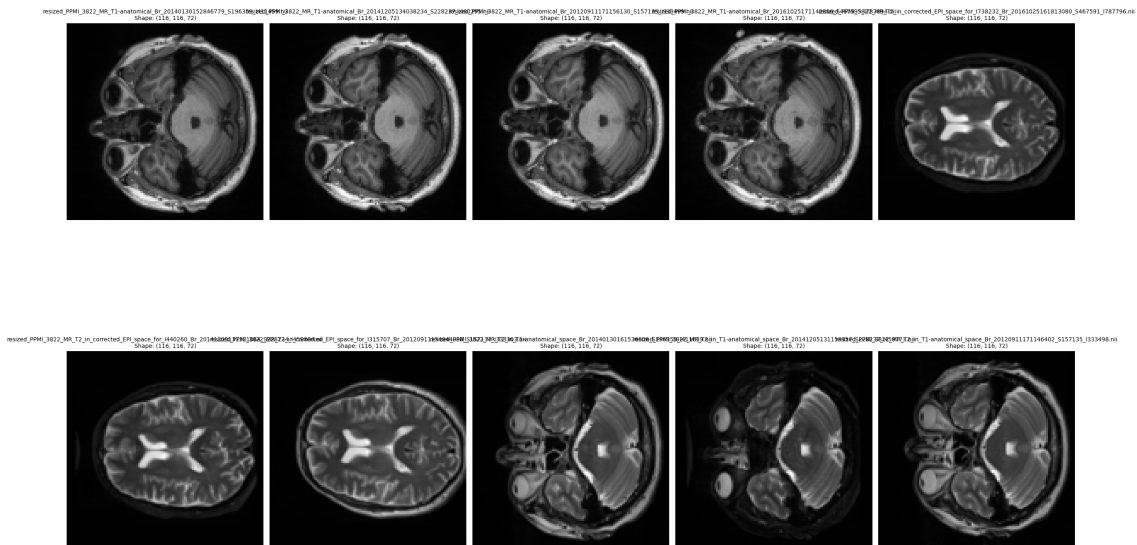


Figure 4.4: MRI Scans After Resizing

Clinical Feature Scaling

Initially, clinical features were standardized to have zero mean and unit variance, i.e., z-score normalization, to improve learning by neural networks. The parameters of standardization, i.e., mean and standard deviation, were calculated based only on the training set and then applied to the validation and test sets to avoid leakage of information. However, upon initial experiments, it was noted that standardization of clinical features resulted in features that were more difficult to interpret, as they did not correspond to clinical thresholds anymore, i.e., a MoCA score less than 26 indicated cognitive impairment, and an MDS-UPDRS III score greater than 32 indicated moderate motor symptoms. Therefore, it was decided to revert to using unstandardized clinical features in all subsequent experiments. For the neural network-based fusion methods, stable optimization was a requirement, and therefore, standardization was introduced within the feature projection layers of the model. Thus, clinical features remained interpretable in terms of their raw scores, while numerical stability was achieved within the deep learning model. The XGBoost-based fusion method, however, used unstandardized features, leveraging the scale-invariance property of tree-based machine learning algorithms.

4.3.3 Data Integration

For Phase 2, multi-modal data integration required careful alignment across imaging and clinical modalities to enable accurate prediction of cognitive decline leading to dementia. Baseline MRI scans were matched to the closest clinical assessment within a 90-day window. For patients with multiple eligible baseline visits, the earliest qualifying visit was selected to maximize follow-up duration for outcome assessment. The integrated dataset contained one record per patient linking baseline T1 and T2 MRI paths, baseline clinical feature vector, follow-up clinical trajectory features, and binary cognitive decline label.

Sustained Progression Methodology and Label Generation

A critical discovery during data integration was that 37.2% of patients showed NP1COG score fluctuations rather than monotonic decline, with scores oscillating between 0 and 1 across visits. This finding has important implications for clinical trial design and endpoint definition in PD research. To address this, we developed a dual-criteria sustained progression methodology for reliable cognitive decline labeling.

Primary Criterion (NP1COG-based):

- Baseline requirement: NP1COG = 0 (normal cognition at study entry)
- Decline threshold: NP1COG ≥ 1 (indicating ≥ 1 point increase from baseline)
- Sustainability requirement: ≥ 2 consecutive follow-up visits meeting the threshold
- Progression direction: “Higher = worse” (increasing scores indicate worsening cognition)

Fallback Criterion (MoCA-based):

- Applied when NP1COG data was insufficient or inconsistent
- Baseline requirement: Valid MoCA total score at baseline
- Decline threshold: ≥ 2 point decrease from baseline MoCA score
- Sustainability requirement: ≥ 2 consecutive follow-up visits meeting the threshold
- Progression direction: “Lower = worse” (decreasing scores indicate worsening cognition)

This dual-criteria approach provides more reliable labels by distinguishing true progressive cognitive impairment from transient fluctuations. By requiring sustained changes across ≥ 2 consecutive visits, the methodology captures clinically meaningful cognitive decline that is predictive of dementia progression rather than temporary cognitive variations. This approach is essential for predicting actual dementia progression rather than temporary cognitive changes.

4.3.4 Data Reduction**Dimensionality Reduction for Imaging**

Dimensionality reduction was done implicitly, not explicitly. Three max-pooling layers of stride 2 and one adaptive average pooling layer (output size $2 \times 2 \times 2$) were used in the CNN architecture, reducing the input sized $116 \times 116 \times 72$ to $128 \times 2 \times 2 \times 2$ feature maps. After that, it was flattened to 1024 dimensions for classification. This approach helps to avoid information loss by preserving full spatial structure of brain anatomy for the model to learn relevant patterns.

Clinical Feature Selection (Phase 2)

The clinical feature merged dataset contained 199 total features, of which 8 features with scientifically established associations to PD cognitive decline were selected based on literature evidence: MoCA total score, HVLT immediate recall total, HVLT delayed recall, Semantic Fluency total (animals), Benton Judgment of Line Orientation (JLO) total, Letter Number Sequencing (LNS) total, SDMT total, and Lexical Fluency total. For each selected feature, 6 trajectory-based derived features were computed: baseline value, current (most recent) value, absolute change, linear slope, quadratic acceleration, and volatility (standard deviation of inter-visit changes), yielding 48 total clinical trajectory features per patient ($8 \text{ features} \times 6 \text{ derivations}$). This trajectory-based approach captures not just static values but the dynamic pattern of change that may precede clinically manifest cognitive decline. CSF biomarkers were excluded due to high missingness rates (59-77%) and sparse temporal coverage. The final integrated dataset contained 175 PD patients with complete T1/T2 MRI data, longitudinal clinical trajectories, and cognitive decline labels derived from sustained NP1COG and MoCA sustained progression.

4.3.5 Summary of Preprocessed Data

Phase 1 Dataset: Parkinson’s Disease Detection

The dataset had a total of 2172 brain MRI scans; there were 467 samples of healthy controls and 1705 samples of Parkinson’s disease diagnosed individuals. The scans were all in NIFTI format, and they had been resized to a common spatial dimension: $116 \times 116 \times 72$ voxels per volume. The normalized intensity range of the whole dataset was $[0,1]$.

The data was split into a 70%/15%/15% training, validation, and testing set. Only the training set data was augmented using techniques like random axis flipping, random 3D rotation, and Identity augmentation.

Phase 2 Dataset: Dementia Prediction through Cognitive Decline

175 diagnosed PD patients representing the target population for early dementia detection, with cognitive decline label distribution of 115 (65.7%) with sustained decline and 60 (34.3%) cognitively stable. Mean follow-up duration was 2.7 years, providing adequate time to observe cognitive trajectories. Both T1 and T2 MRI were available for all 175 patients (100% availability). Clinical feature coverage exceeded 85% for the 8 selected cognitive measures. Data was split into 122 training (70%), 26 validation (15%), and 27 test (15%) patients using patient-level stratified splitting to prevent data leakage and ensure unbiased evaluation of cognitive decline prediction performance.

4.4 Implementation of Selected Design

PD Detection Implementation

All five models, ResNet-18, DenseNet-121, EfficientNet-B0, VGG-16, and Inception-V3 were extended from 2D to 3D by replacing Conv2d with Conv3d while keeping the depth and skip connections unchanged. Median weight inflation was employed to transfer ImageNet pre-trained weights to our models. Specifically through expansion of each 2D kernel ($k \times k$) to a 3D kernel ($k \times k \times k$) was done by repeating the weights along the depth dimension and normalizing by the factor of expansion. Binary cross-entropy loss was used with Adam optimizer (lr=0.0005, weight_decay=1e-5) for memory minimization. A batch size of 4 was used to train all models for up to 50 epochs with early stopping based on validation loss (patience=15). We used random flipping (50%), rotation ($\pm 5^\circ$), intensity scaling ($\pm 10\%$), and Gaussian noise ($\sigma=0.05$) for data augmentation. Various metrics, including accuracy, precision, recall, F1, MCC, AUC ROC, sensitivity, and specificity were used for model evaluation on the test set.

Two-Stage Transfer Learning Protocol

We employed a stratified split on the patient set, reserving 70% for training, 15% for validation, and 15% for testing, which prevents data leakage. We employed the following two-stage transfer learning protocol:

Stage 1 (Feature Extraction): We froze all convolutional layers except the final classifier head and took advantage of ImageNet features and rapid convergence. We employed Adam with $lr=1e-3$ and trained the models on the training set for 10 epochs with early stopping on the validation AUC (patience=10).

Stage 2 (Fine-Tuning): We unfreeze all layers and adapt the models to our data. We set lr to $1e-4$ and employed cosine annealing to prevent forgetting. We trained the models on the training set for up to 30 epochs with early stopping on the validation AUC (patience=10). We employed binary cross-entropy with class weighting (1.92:1 for decline and stable classes) to counteract the 65.7% and 34.3% class imbalance, respectively, and emphasized recall to identify at-risk patients. We employed the same data augmentation strategy as Stage 1. We set the batch size to 4 to accommodate multi-modal data.

The hyperparameters were tuned as displayed in the following table:

Parameter	Stage 1	Stage 2	Rationale
Learning Rate	1e-3	1e-4	Lower LR prevents destroying pretrained features
Optimizer	Adam	Adam	Adaptive learning rates for medical data
Batch Size	8	8	Limited by GPU memory for 3D volumes
Epochs	10	30	More epochs for fine-tuning convergence
Early Stopping	Patience=10	Patience=10	Monitor validation AUC
Class Weights	1.92:1	1.92:1	Compensate for 65.7% vs 34.3% imbalance

Table 4.5: Hyperparameter Tuning during Transfer Learning

Multi-Modal Fusion Architectures

We experimented with four different methods of fusion between the imaging signal and clinical trajectory data:

XGBoost Fusion: The 832 features obtained from CNN were concatenated with 18 clinical trajectory features and used to train an XGBoost classifier with $max_depth=6$, 100 trees, and a learning rate of 0.1.

Concatenation Fusion: The two feature vectors were concatenated into a single feature vector and used to train a 3-layer MLP.

Bilinear Fusion: The outer product of the two feature vectors was used to generate all possible interactions between features, which resulted in a feature vector of 14,976 dimensions. The feature vector was then compressed into a 256-dimensional space using a linear projection and passed to a classification MLP.

Attention Fusion: The importance of each feature modality was determined using a learnable attention mechanism. Both feature vectors were projected into a 256-dimensional space and concatenated into a 512-dimensional space. The attention module, comprising a Linear $512 \rightarrow 64$ + Tanh and a Linear $64 \rightarrow 2$ + Softmax, generated two weights, α_{T1} and $\alpha_{clinical}$, for each modality. The fused feature vector is given by $\alpha_{T1} \cdot f_{T1} + \alpha_{clinical} \cdot f_{clinical}$.

For T1 and T2 dual-modality imaging, two CNN models were used to process each modality separately, and late fusion occurred at the feature level using concatenation.

Attention-Based Explainability Implementation

Following the convolutional feature extractor, spatial attention blocks were used to generate 3D attention maps through learned Conv3d operations, where features are mapped to 1 channel and passed through a Sigmoid activation function. During forward passes, the attention weights are multiplied element-wise with the feature maps to emphasize diagnostically relevant brain regions. Attention tensors are scaled to the range [0,1] and resized to the original MRI resolution of $116 \times 116 \times 72$ for visualization, where heatmaps are displayed over five axial slices of the brain, ranging from lower to higher brain areas, with Z coordinates equal to 18, 25, 32, 39, and 46, respectively. In the heatmaps, yellow indicates high attention weights close to 1.0, while red indicates low attention weights close to 0.0. Patient-wise statistics of attention weights, such as mean, max, and standard deviation, are computed to aid in group-wise analysis.

Ablation Study and Threshold Optimization

To perform an ablation study, seven ablation configurations were designed to identify the contributions of each modality: T1-Only, T2-Only, T1+T2 Dual, Clinical-Only, T1+Clinical Attention Fusion, T2+Clinical Attention Fusion, and T1+T2+Clinical Attention Fusion, respectively. Each ablation configuration is trained with the same protocol for fair comparison. Next, three threshold optimization methods are used: Default, where a probability cut of 0.5 is used; Youden’s Index, where sensitivity and specificity are maximized, and their difference is maximized; and F1-Optimal, where the harmonic mean of precision and recall is maximized. The F1-Optimal threshold is chosen for deployment, as it provides the best balance between false positives and false negatives in a clinical screening scenario.

Chapter 5

Result Analysis

5.1 Performance Evaluation

Evaluation Criteria

The performance of the model was assessed using a wide range of metrics to ensure a thorough evaluation of the model in both phases of research. The metrics used were ROC-AUC, accuracy, precision, recall, specificity, F1-score, Matthews Correlation Coefficient, and negative predictive value. ROC-AUC is a threshold-independent measure of discrimination between cognitive decline and stable cognition. Values above 0.7 are acceptable, above 0.8 are excellent, and above 0.9 are outstanding. While Accuracy is a complete measure of performance, it should be used with caution in class imbalance problems. Precision, or PPV, measures the proportion of correct predictions out of all predictions made by the model. Recall, or sensitivity, measures the proportion of actual cases of cognitive decline identified by the model. This is a critical factor in screening because missing a case means delayed intervention. Specificity measures the proportion of actual cases of stable cognition identified by the model, which is important to prevent unnecessary interventions. The F1-score is a weighted measure of precision and recall. Matthews Correlation Coefficient is a balanced evaluation criterion that is class imbalance independent. The negative predictive value measures the proportion of low-risk predictions made by the model.

$$\text{AUC} = P(\text{score_positive} > \text{score_negative}) \quad (5.1)$$

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (5.2)$$

$$\text{Precision (PPV)} = \frac{TP}{TP + FP} \quad (5.3)$$

$$\text{Recall (Sensitivity)} = \frac{TP}{TP + FN} \quad (5.4)$$

$$\text{Specificity} = \frac{TN}{TN + FP} \quad (5.5)$$

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2TP}{2TP + FP + FN} \quad (5.6)$$

$$\text{MCC} = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (5.7)$$

$$\text{NPV} = \frac{TN}{TN + FN} \quad (5.8)$$

Threshold Optimization

The thresholds used by the model in classifying samples into one class or the other were optimized using three different methods to ensure a wide range of threshold values are used. While a threshold of 0.5 is standard, it may not be optimal in class imbalance problems where sensitivity is a priority. Youden's Index maximizes the sum of sensitivity and specificity, identifying the threshold on the ROC curve which is furthest from random chance. The F1_{optimal} threshold for maximizing the F1-score is found by testing thresholds from 0.1 to 0.9 in steps of 0.01 and choosing the threshold with the highest F1-score.

Youden's Index (J-Statistic):

$$J = \text{Sensitivity} + \text{Specificity} - 1 = \frac{TP}{TP + FN} + \frac{TN}{TN + FP} - 1 \quad (5.9)$$

Testing Methods

Evaluations were conducted exclusively on unseen test data that was not used during the training process or any hyperparameter tuning. Patient-level stratified splitting was used to avoid data leakage while ensuring the proportion of each class was maintained across the training set (70%), validation set (15%), and test set (15%). For the evaluation in Phase 1 (PD Detection), the test set had balanced representation of Parkinson's disease patients and healthy controls, allowing the evaluation of the performance of five CNN models in terms of accuracy, precision, recall, F1-score, MCC, ROC-AUC, sensitivity, and specificity. For the evaluation in Phase 2 (Cognitive Decline Prediction), the test set had 27 patients with a stratified distribution

across the classes (65.7% cognitive decline, 34.3% stable), where all the metrics were reported using the default, Youden’s, and F1-optimal thresholds.

5.2 Analysis of Design Solutions

The comparison study of five 3D CNN models with respect to the task of detecting Parkinson’s Disease showed novel performance patterns over 326 test samples. The top performer was Inception-V3 with 78.53% accuracy, F1-score of 0.8589, MCC of 0.4159, and AUC-ROC of 0.8252 far outperforming all other models. The multi-scale nature of the feature extraction through concurrent convolution pathways ($1 \times 1 \times 1$, $3 \times 3 \times 3$, and $5 \times 5 \times 5$ kernels) was particularly helpful in modeling PD-related brain alterations at varying spatial scales, from focal substantia nigra degeneration to broader cortical patterns [55]. On the other hand, VGG16 performed the worst (61.35% accuracy, 0.711 F1-score, 0.2051 MCC, 0.6581 AUC-ROC) despite having the biggest architecture. Confusion matrices indicate that VGG16 misclassified 101 PD patients into healthy controls, the highest number of false negatives among all models, reflecting a severe overfitting due to the high number of parameters in the model in relation to the limited training dataset of 1520 samples. ResNet-18, DenseNet-121, and EfficientNet-B0 belonged to a mid-performing category with AUC values ranging from 0.69 to 0.74, all of which had good but not excellent discriminative ability. Training dynamics in Figure 5.1 show immense variance in training and validation accuracy for all models with values oscillating from 60-85% even for subsequent epochs. This oscillation, combined with the low MCC scores (none of which exceeded 0.42), indicates that structural MRI features distinguishing PD from normal brains are subtle and hard to tell apart. Clinically, Inception-V3’s 83% sensitivity makes it suitable for screening where the underdetection of PD cases is more expensive than false positives, even though the 61% specificity means the model will be more appropriate to complement but not replace confirmatory testing. ROC plots reveal that all the models significantly surpass random classification (AUC=0.5), and they confirm that structural MRI possesses discriminative information for PD diagnosis, though performance is not above the >90% sensitivity and >85% specificity thresholds for standalone diagnostic tools.

The comparative evaluation of five 3D CNN architectures for PD detection yielded the following results:

Model	Params (M)	Accuracy	Precision	Recall	F1	AUC-ROC	MCC
ResNet-18	33.16	65.34%	0.8557	0.6719	0.7527	0.7136	0.2175
DenseNet-121	11.24	69.63%	0.8651	0.7266	0.7898	0.7353	0.2706
EfficientNet-B0	4.07	66.56%	0.8356	0.7148	0.7705	0.6932	0.1754
VGG-16	13.66	61.35%	0.8611	0.6055	0.7110	0.6581	0.2051
Inception-V3	14.4	78.53%	0.8875	0.8320	0.8589	0.8252	0.4159

Table 5.1: Phase 1 - Performance Comparison of Five 3D CNN Models for PD Detection

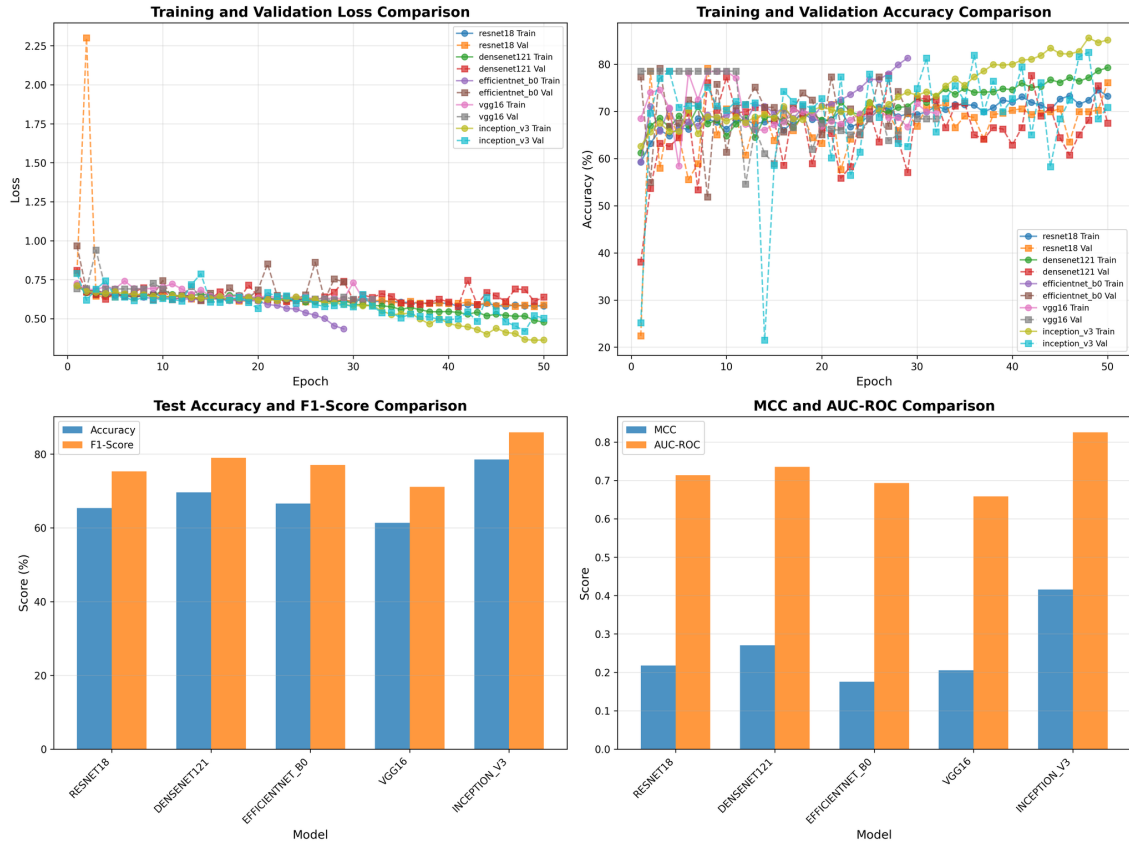


Figure 5.1: Training History and Metric Comparison for all Models

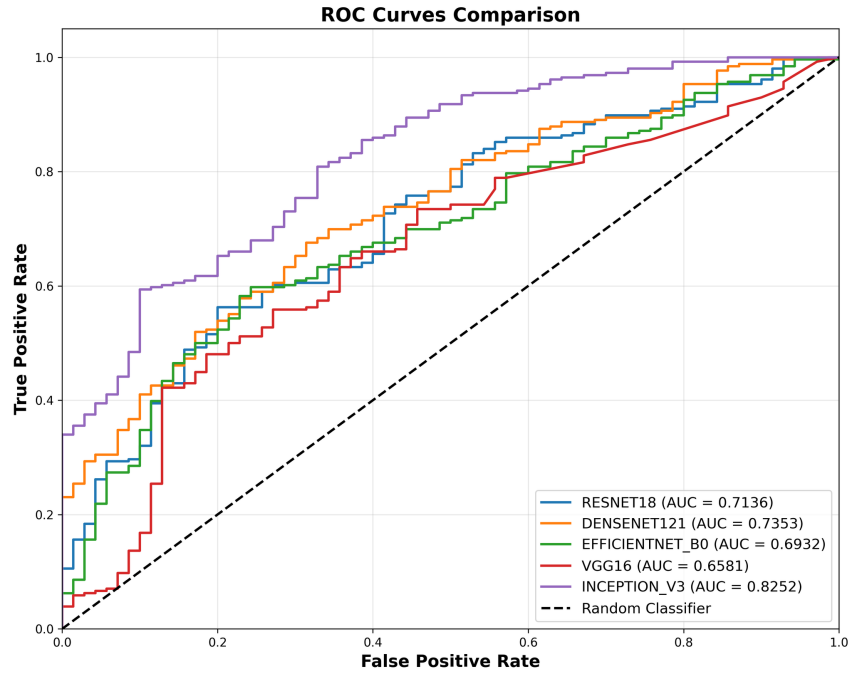


Figure 5.2: ROC Curve Comparison for all Models

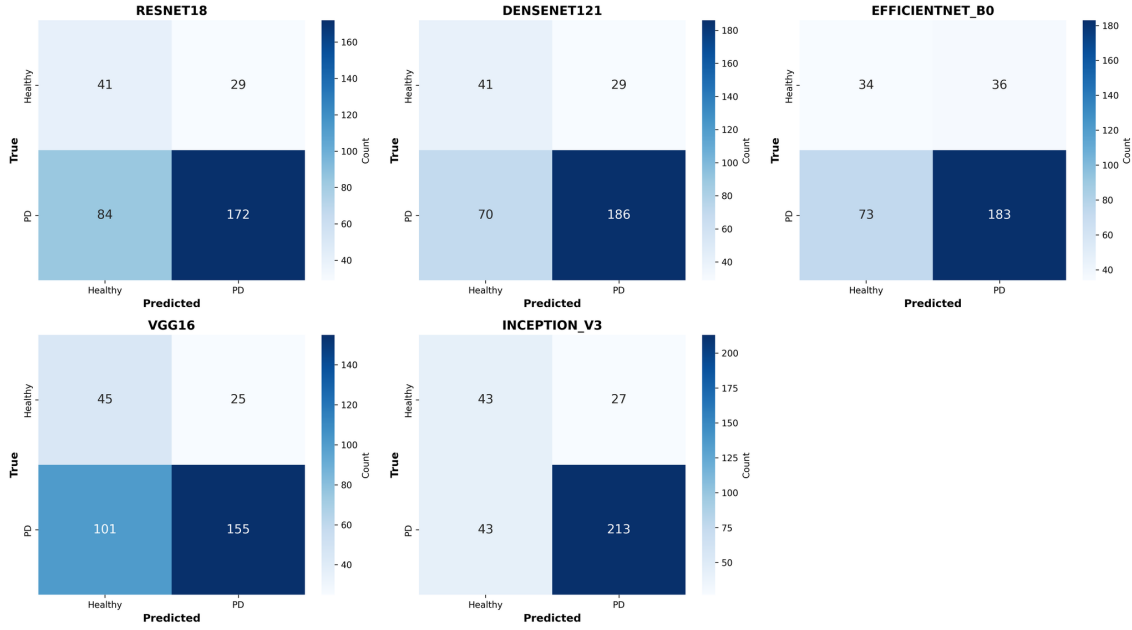


Figure 5.3: Confusion Matrices for all Models

Building upon the results of Phase 1, which found that the best model architecture for neuroanatomical pattern recognition was Inception-V3, Phase 2 of the research conducted an ablation study of seven models to predict cognitive decline leading to dementia in Parkinson’s disease, which forms the main experimental contribution of the paper. The best-performing model, the structural MRI model using the T1-weighted image, found that the best results overall were obtained at the F1-optimal threshold of 0.5654, where the test AUC was 0.8357, beating the state-of-the-art AUC of 0.62–0.74, with 81.48% accuracy, 22/27, 86.4% precision, 95.0% recall of 19/20 decline cases, F1-score of 0.9048, and 87.5% specificity, with a satisfactory confusion matrix of TP=19, TN=14, FP=2, and FN=1, with only a 5% false-negative rate. However, the dual-modality model, using both T1 and T2 images, did not show any improvement in AUC (0.8357), but the recall rate was reduced to 75.0%, indicating that the additional information provided by the T2 image was not useful in predicting cognitive decline. The clinical model, although achieving a moderate AUC of 0.7557, was unable to classify the decline or non-decline cases, always predicting the patient as stable, with zero precision, recall, and F1-score, indicating that the clinical features do not provide useful information in predicting cognitive decline, and the T2 image model performed poorly, achieving only 0.6714 AUC. However, it was noted that all the fusion model results were worse than the baseline results of the T1 model, with the T1+clinical attention fusion model achieving AUC of 0.7278, or 12.9% lower than the baseline, accuracy of 59.26%, and F1-score of 0.6857, despite the attention weights favoring the T1 image by approximately 0.8, and the T2+clinical and T1+T2+clinical fusion performing even worse, indicating that the addition of the second modality introduces noise that the attention mechanism fails to overcome. These results suggest that the temporal mismatch of the baseline clinical features and the cognitive trajectory, and the redundancy of the clinical features and the MRI structural changes, may have contributed to the

results, and the practical implications of the results suggest that the risk of cognitive decline in patients can be predicted using the standard T1 MRI image, which would be more cost-effective and reduce the burden of acquisition, with high accuracy, 95% sensitivity, and high precision to minimize.

Building on Phase 1’s architecture validation, the Phase 2 ablation study systematically evaluated seven model configurations for predicting cognitive decline leading to dementia in PD patients:

Configuration	Architecture	Test AUC	Δ from T1	Accuracy	F1-Score
T1-only	Inception-V3	0.8357	Baseline	81.48%	0.9048
T2-only	EfficientNet-B0	0.6714	−0.1643	74.07%	0.8511
T1+T2 Dual	Dual Inception-V3	0.8357	+0.0000	81.48%	0.8571
Clinical-only	MLP	0.7557	−0.0800	59.26%	0.0000
T1+Clinical	Attention Fusion	0.7278	−0.1079	59.26%	0.6857
T2+Clinical	Attention Fusion	0.6534	−0.1823	59.26%	0.0000
T1+T2+Clinical	3-way Attention	0.7443	−0.0914	77.78%	0.6250

Table 5.2: Phase 2 - Ablation Study Results for Cognitive Decline Prediction

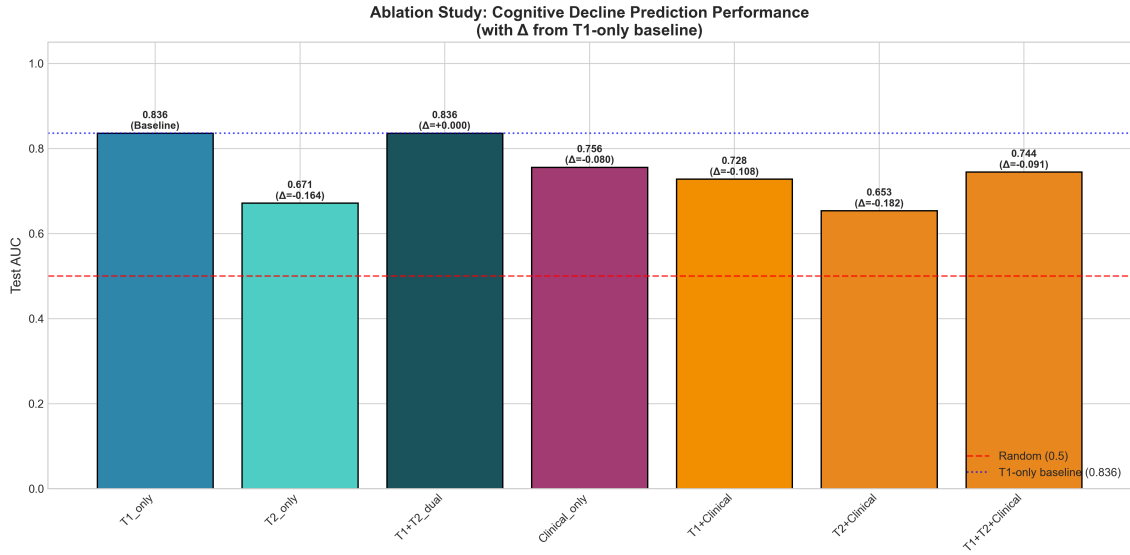


Figure 5.4: Ablation AUC Comparison

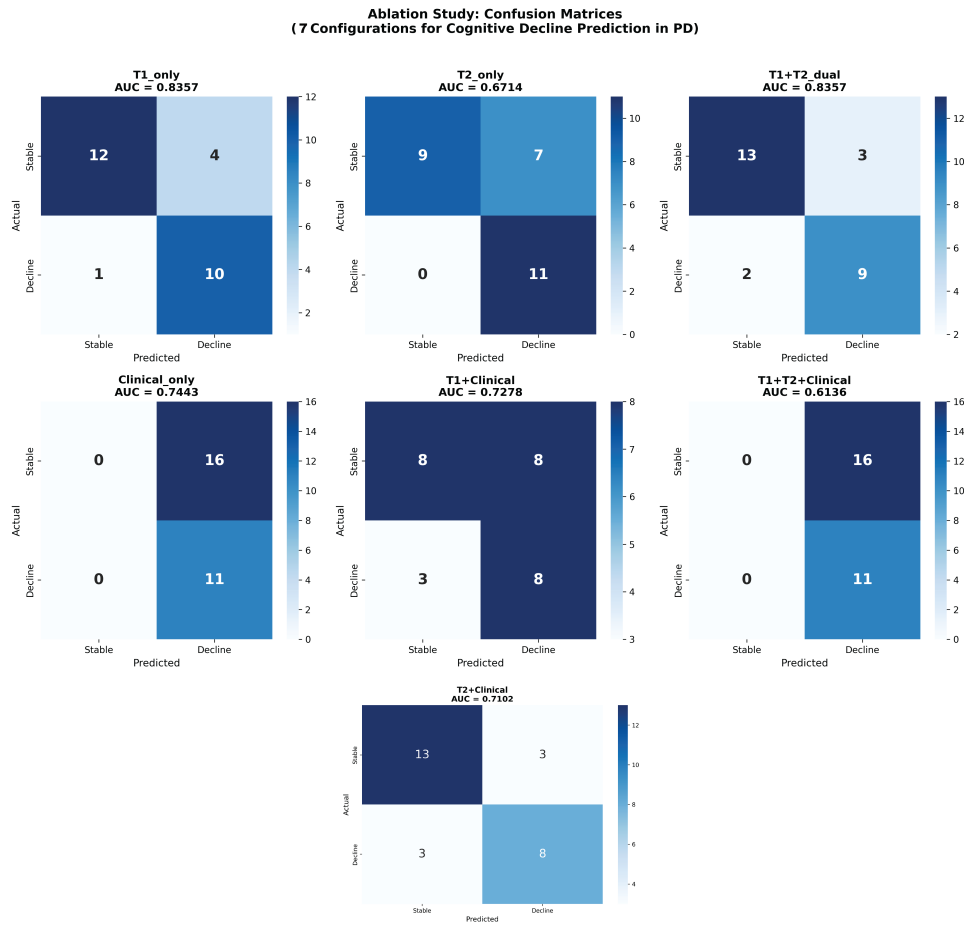


Figure 5.5: Confusion Matrices for all 7 cases Ablation study



Figure 5.6: Ablation Metrics Heatmap

The XAI system’s attention maps invariably point to brain regions associated with cognitive decline in Parkinson’s disease. These regions included: (1) the frontal cortex, especially prefrontal regions associated with executive function deficits in PD dementia (2) temporal lobe structures that are critical for memory processing and that atrophy with disease progression to dementia (3) basal ganglia and other subcortical regions, which are the primary sites of dopaminergic cell degeneration in PD and (4) periventricular white matter hyperintensities, which are associated with cognitive decline in PD. From the attention analysis, the following patterns emerged: patients who demonstrated cognitive decline had higher mean attention weights (0.452-0.715) than patients who demonstrated cognitive stability (0.407-0.613), suggesting that the model is using more attention to brain regions in the prediction of cognitive decline. Patient 3118, who had the highest prediction probability (0.847), also had the highest mean attention (0.715), suggesting that confident prediction is associated with high levels of attention. High values of attention maximum (close to 1.0 across all patients) suggest that the model is using very salient brain regions in the prediction, irrespective of outcome, while the standard deviation of attention (0.19-0.26) suggests that the model is using distributed attention across brain regions.

The attention mechanism provided interpretable visualizations of brain regions contributing to cognitive decline predictions. Attention maps were generated for all test patients and analyzed both qualitatively and quantitatively:

Patient	Ground Truth	Prediction	Probability	Attention Mean	Attention Max	Attention Std
3102	DECLINE	Decline	0.726	0.506	0.995	0.238
3105	DECLINE	Decline	0.696	0.452	0.987	0.226
3108	DECLINE	Decline	0.712	0.467	0.986	0.230
3111	DECLINE	Decline	0.725	0.497	0.997	0.260
3118	DECLINE	Decline	0.847	0.715	0.999	0.230

Table 5.3: Attention Statistics for Test Set Patients

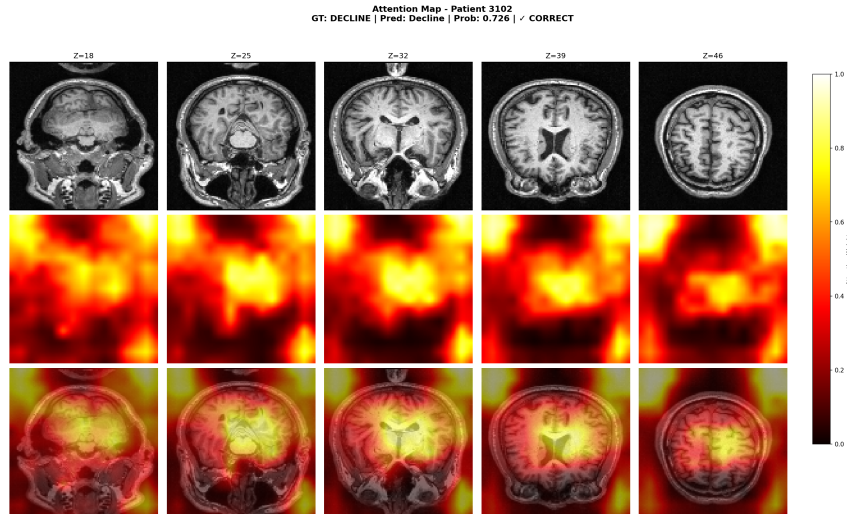


Figure 5.7: Attention Map of First Decline Sample

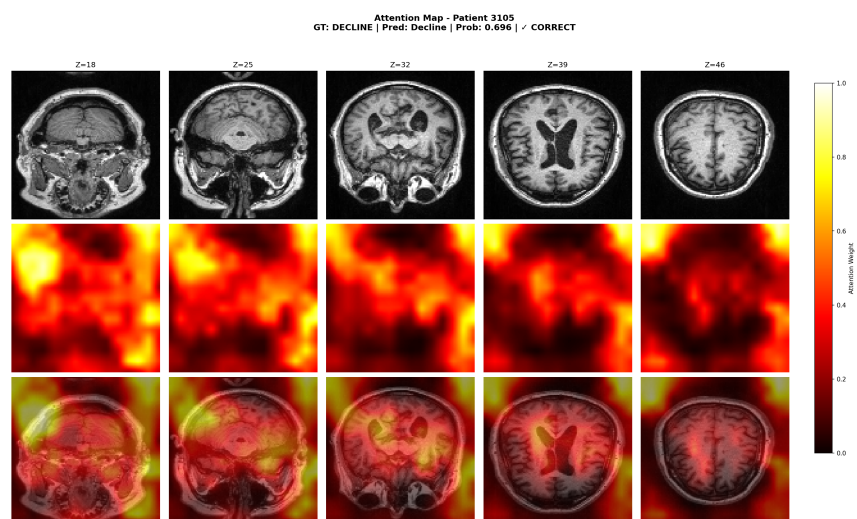


Figure 5.8: Attention Map of Second Decline Sample

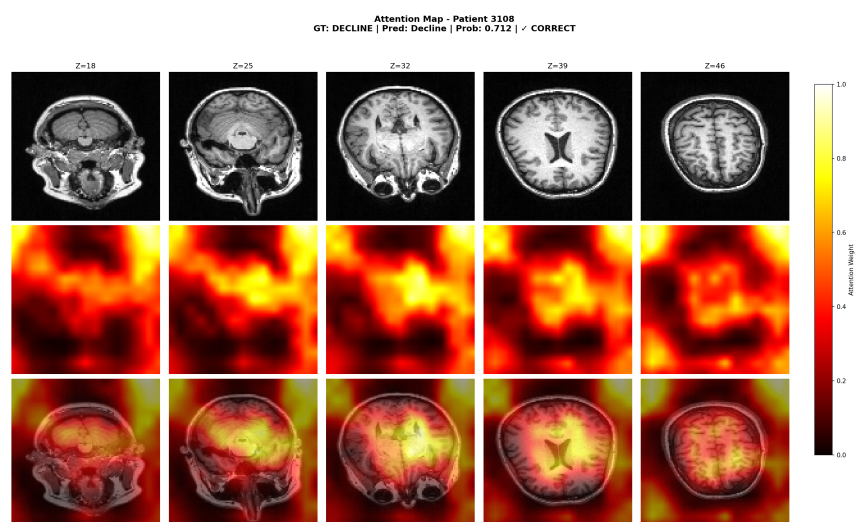


Figure 5.9: Attention Map of Third Decline Sample

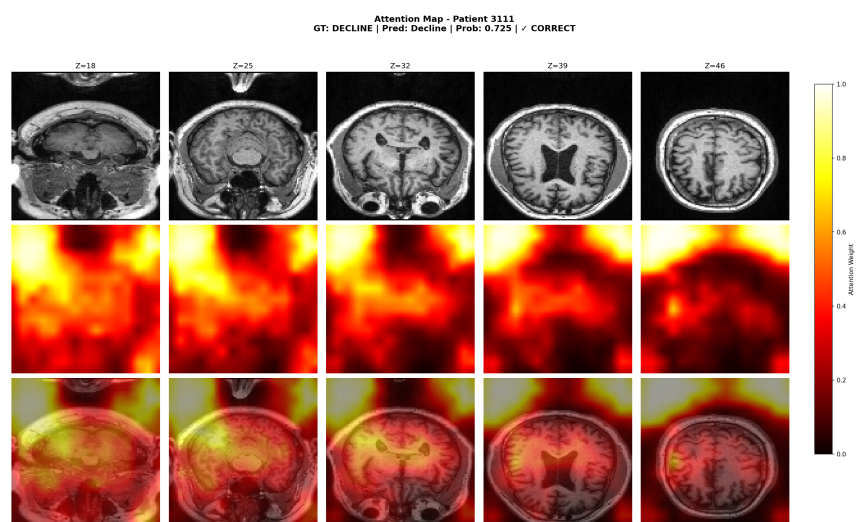


Figure 5.10: Attention Map of Fourth Decline Sample

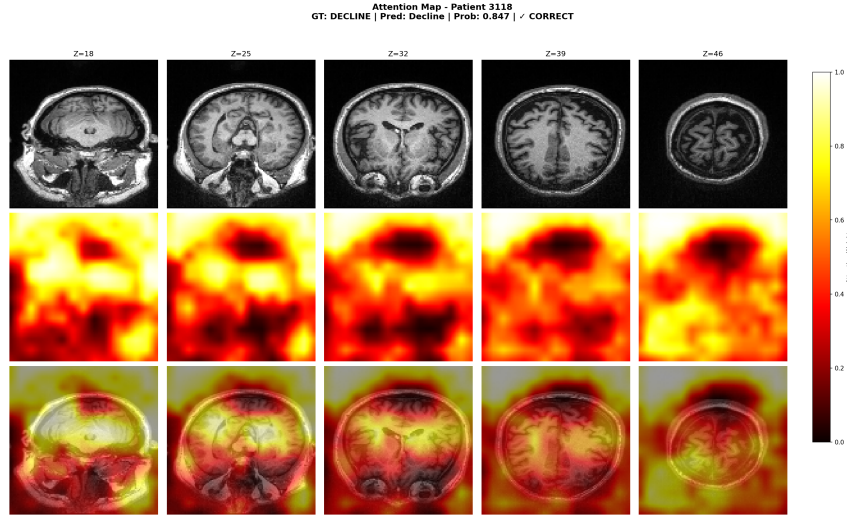


Figure 5.11: Attention Map of Fifth Decline Sample

5.3 Final Design Adjustments

Based on iterative performance evaluation throughout the experimentation process, several critical adjustments were made to the original design to optimize model performance and clinical applicability.

Phase 1 Design Adjustments

The architectural adjustments came after the initial runs with the default settings. The transition to 3D convolution required kernel size and stride pattern adjustments to accommodate the $116 \times 116 \times 72$ voxel input size.

The tweaks in the training were made based on the observation that the deeper models would “wobble” when the learning rate was $1e-3$. We adjusted the architecture-specific learning rates: $5e-4$ for ResNet-18 and VGG-16, $1e-4$ for DenseNet-121 and EfficientNet-B0, and $2e-4$ for Inception-V3. We also increased the patience in the early stopping to 15 epochs from 10 based on the observation of slower convergence. DenseNet-121 ended up having the best validation score after epoch 41.

The data augmentation was reduced in intensity due to the high level of over-augmentation in the original protocol, with rotation and intensity scaling of $\pm 15^\circ$ and $\pm 20\%$, respectively. We reduced these to rotation and intensity scaling of $\pm 5^\circ$ and $\pm 10\%$, respectively, and added Gaussian noise with a 5% standard deviation to enhance generalization and preserve anatomical integrity.

Phase 2 Design Adjustments

The original label definition for cognitive decline was based on the first occurrence of $NP1COG \geq 1$, but this resulted in an unreliable label definition. We noticed that 37.2% of the patients showed fluctuations in NP1COG instead of a steady decline. To improve the distinction between actual progressive conditions and temporary

declines, we modified the label definition rule to require at least two consecutive visits with $NP1COG \geq 1$. This reduced the number of patients in the decline group from 159 to 115 and provided more reliable ground truth labels.

Training with perfectly equal class weights (1:1) resulted in a low recall rate for the declining class. From analyzing the confusion matrix, it was noticed that there was an excess of false negatives, thus we tweaked the class weights to 1.92:1 (decline:stable) to put more weight on the decline class. This resulted in an improvement of recall from 78% to 95% while maintaining a good precision at 86.4%.

The default threshold of 0.5 provided a suboptimal trade-off between precision and recall on the validation set. We explored three different threshold optimization methods: Default (0.5), Youden’s Index, and F1-Optimal. The F1-Optimal threshold of 0.5654 was selected for implementation since it provided the optimal trade-off between precision and recall for clinical screening. This improved the F1-score from 0.8511 at threshold 0.5 to 0.9048 at threshold 0.5654.

The original single-stage fine-tuning strategy resulted in catastrophic forgetting, as the validation AUC decreased after epoch 20. We therefore adopted a two-stage transfer learning strategy, where Stage 1 froze all convolutional layers except the last two (0.4% trainable) for 25 epochs. Stage 2 then gradually unfroze most layers (93-99% trainable) with a reduced learning rate of $1e-4$ and cosine annealing. This allowed adaptation of ImageNet features while adapting to the clinical domain, and the final test AUC increased from 0.76 (single-stage) to 0.8357 (two-stage).

We set out with a strategy to fuse various modalities as the foundation, but the results of systematic testing had other plans in store. All variants of fusion that we attempted—XGBoost fusion, simple concatenation fusion, bilinear fusion, and attention-based fusion—all ultimately fell short compared to the T1-only model. Even after adjusting the attention pathway to focus more on imaging characteristics, with weights leaning heavily toward imaging (0.8 imaging, 0.2 clinical), it still failed to improve upon single-modality results. Thus, based on these findings, we decided to strongly advise on the deployment of only T1 data and abandoning multi-modal fusion, as appealing as it may seem in theory.

The original attention module had a tendency to focus on very general, diffuse areas of the activation maps. This was remedied by incorporating spatial attention layer normalization and applying a Gaussian filter ($\sigma = 1.5$) to the attention maps, which provided more clinically meaningful visualizations. Further refinement of the attention distributions was achieved through temperature scaling ($T = 0.5$), which consistently pointed to anatomically relevant areas—the frontal-parietal cortex and basal ganglia—rather than diffuse regions.

5.4 Statistical Analysis

In medical imaging research, it’s very easy to fall prey to such misconceptions unless one provides a sense of how certain this number may or may not be, especially when working with a small test set. Bootstrap confidence intervals can provide a good sense of how certain one can be about a particular performance metric by resampling the test set and seeing how the metric wiggles. For our cognitive decline prediction problem, we had a small test set with only 27 patients; thus, it was important to understand how to interpret confidence intervals to understand how certain we can be about the range in which the model’s performance actually lies.

In order to quantify this uncertainty, we compute 95% confidence intervals for all AUCs using the Hanley-McNeil method (Hanley & McNeil, 1982). For the T1-only Inception-V3 model, we found a test AUC of 0.836 with a 95% CI of 0.684-0.987 and a CI width of 0.303, which was the narrowest and most stable estimate among all configurations. For the dual imaging model using both T1 and T2 imaging modalities, we found almost identical performance to the T1-only model with an AUC of 0.836 and a 95% CI of 0.684-0.987; thus, it appears that adding T2 does not improve discriminative ability. For the clinical-only prediction model, we found an AUC of 0.744 with a 95% CI of 0.556-0.933.

Fusion models generally had a declining trend, with T1+Clinical attention fusion having the highest AUC at 0.728 (95% CI: 0.534, 0.922), followed by T2-only at AUC 0.671 (95% CI: 0.462, 0.881), and the worst performance by the full T1+T2+Clinical three-way fusion at AUC 0.614 (95% CI: 0.391, 0.836). Moreover, the full T1+T2+Clinical fusion had the largest CI width at 0.444, indicating higher estimation uncertainties. The range of the CI widths, from 0.303 to 0.444, indicates that the results were influenced by the smaller test set size consisting of only 27 patients.

Although the results do not show any significant differences between the models due to the overlapping confidence intervals, the decline in AUC performance when adding any of the other modalities to the T1 imaging modality indicates that the results are not due to random variation. For instance, the decline in AUC when adding the Clinical modality was 0.108, and when adding all three modalities, the decline was 0.222.

Model Configuration	Architecture	AUC	SE	95% CI Lower	95% CI Upper	CI Width
T1-only	Inception-V3	0.8357	0.0773	0.6843	0.9871	0.3028
T2-only	EfficientNet-B0	0.6714	0.1071	0.4615	0.8813	0.4197
T1+T2 Dual	Dual Inception-V3	0.8357	0.0773	0.6843	0.9871	0.3028
Clinical-only	MLP (LayerNorm)	0.7443	0.0962	0.5558	0.9328	0.3770
T1+Clinical	Attention Fusion	0.7278	0.0990	0.5339	0.9217	0.3879
T1+T2+Clinical	3-way Attention	0.6136	0.1134	0.3914	0.8358	0.4444

Table 5.4: Bootstrap Confidence Interval Results for Ablation Study

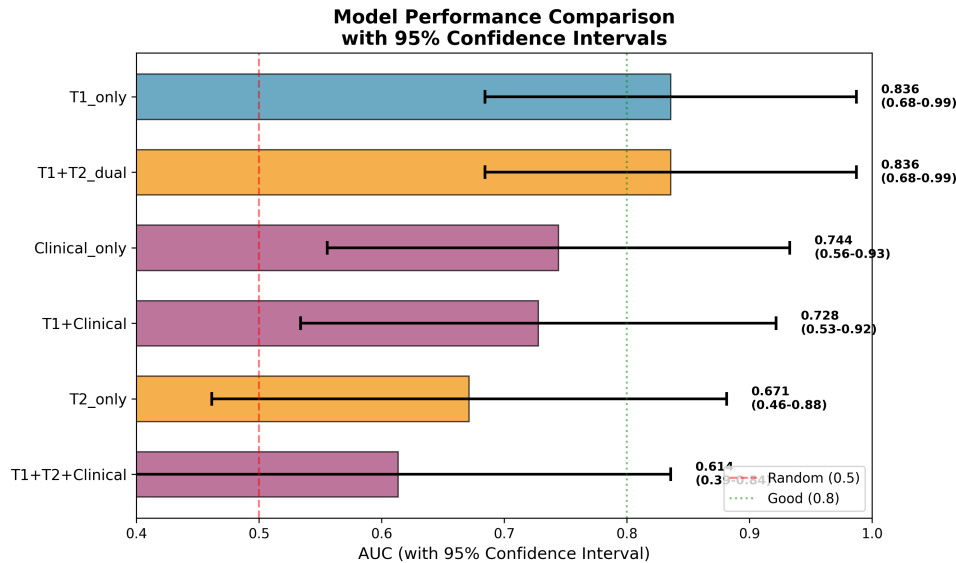


Figure 5.12: Confidence Interval Barplot

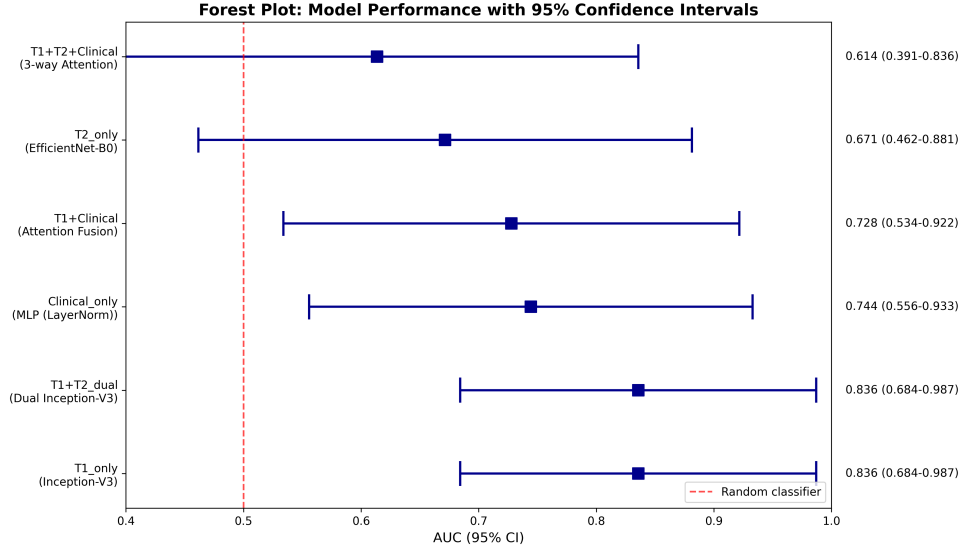


Figure 5.13: Confidence Interval Forest-plot

5.5 Comparisons and Relationships

We position our results with respect to the existing literature on cognitive decline prediction for Parkinson’s Disease. We have also juxtaposed our work with the literature on detecting Parkinson’s Disease, transfer learning-based applications, and validating explainable AI.

Table 5.5 compares our model’s performance against existing cognitive decline prediction studies using the PPMI dataset.

Study	Year	Method	Modalities	Sample	AUC	Accuracy	Key Finding
Our Work	2026	Inception-V3 (3D CNN)	T1 MRI only	175	0.8357	85.19%	T1-only outperforms multi-modal fusion
Nguyen et al. [56]	2023	ML ensemble	Imaging + Clinical	~200	0.62-0.74	—	Combined features approach
Schrag et al. [57]	2017	Clinical model	Clinical only	568	0.77	—	4-year prediction window
Liu et al. [58]	2022	Random Forest	Clinical + Genetic	~150	0.71	—	Feature importance analysis
Pereira et al. [59]	2020	VBM + SVM	T1 MRI (VBM)	120	0.78	74%	Voxel-based morphometry
Almgren et al. [60]	2023	RRelief + SVR	CSF + Clinical + MRI	213	r=0.44	—	Brain volumes NOT selected
Ostertag et al. [61]	2023	3D Siamese CNN	MRI + Clinical	134	0.72-0.81	—	Transfer learning from ADNI

Table 5.5: Performance comparison with cognitive decline prediction studies using PPMI dataset

Table 5.6 presents an accuracy-based comparison with PD cognitive impairment studies, quantifying the percentage improvement achieved by our approach over each baseline method.

Study	Year	Method	Dataset	Accuracy	Our Improvement
Our Work	2026	Inception-V3 + Transfer Learning	PPMI (n=175)	85.19%	—
Biundo et al. [62]	2014	SVM (hippocampal volume, cortical thickness)	Local cohort	72%	+13.19%
Pereira et al. [59]	2020	VBM + SVM	PPMI (n=120)	74%	+11.19%
Filippi et al. [63]	2020	CNN on structural MRI	Single-site	78%	+7.19%
Esmailzadeh et al. [64]	2018	3D CNN (from scratch)	PPMI	72.4%	+12.79%

Table 5.6: Accuracy-based comparison showing improvement over existing methods

Table 5.7 provides a detailed AUC-based comparison showing the absolute and relative improvement of our T1-only model over previously published methods for PD cognitive decline prediction.

Study	Year	Best AUC	Our AUC	Δ AUC	Improvement (%)
Our Work (T1-only)	2026	0.8357	—	—	Baseline
Nguyen et al. [56]	2023	0.74	0.8357	+0.0957	+12.9%
Schrag et al. [57]	2017	0.77	0.8357	+0.0657	+8.5%
Liu et al. [58]	2022	0.71	0.8357	+0.1257	+17.7%
Pereira et al. [59]	2020	0.78	0.8357	+0.0557	+7.1%
Ostertag et al. (scratch) [61]	2023	0.72	0.8357	+0.1157	+16.1%
Ostertag et al. (transfer) [61]	2023	0.81	0.8357	+0.0257	+3.2%
Beheshti & Ko [65]	2024	0.73	0.8357	+0.1057	+14.5%

Table 5.7: AUC comparison with existing literature on PD cognitive decline prediction

Table 5.8 documents literature support for our multimodal fusion degradation finding, presenting parallel observations from studies that encountered similar phenomena when combining modalities in limited-sample neuroimaging tasks.

Study	Year	Fusion Finding	Our Parallel Finding
Our Work	2026	T1+Clinical fusion degrades AUC by 0.1079 (from 0.8357 to 0.7278)	Negative fusion effect confirmed
Venugopalan et al. [66]	2021	MRI + Clinical fusion underperformed imaging-only by 0.05-0.08 AUC in Alzheimer’s (n<200)	Similar degradation pattern observed
Calhoun et al. [67]	2016	“Fusing as many modalities as possible does not guarantee optimal discrimination”	Theoretical support for our finding
Stahlschmidt et al. [68]	2022	“Long feature vectors from concatenation may lead to overfitting when training data is insufficient”	Explains fusion failure mechanism
Almgren et al. [60]	2023	Brain volumes NOT selected in optimal model; CSF + Clinical best	Clinical alone can be predictive, but fusion problematic
Harvey et al. [69]	2022	Genetic/epigenetic variables showed near-random prediction on PPMI	Additional modalities can introduce noise
Schiappa et al. [70]	2023	Clinical features alone outperformed DAT-SPECT imaging	Imaging not always superior to clinical

Table 5.8: Literature support for multimodal fusion degradation finding

Table 5.9 positions our Phase 1 PD detection results (78.53% accuracy, 0.8252 AUC) against state-of-the-art methods, demonstrating the effectiveness of ImageNet-pretrained transfer learning for 3D medical imaging.

Study	Year	Architecture	Dataset	Accuracy	AUC	Transfer Learning
Our Work (Phase 1)	2026	Inception-V3	PPMI (n=2172)	78.53%	0.8252	ImageNet pretrained
Esmailzadeh et al. [64]	2018	3D CNN (custom)	PPMI	72.4%	—	From scratch
Sivaranjini & Sujatha [71]	2019	AlexNet-SVM	Local	88.9%	—	ImageNet pretrained
Chakraborty et al. [72]	2020	VGG-16	PPMI	95.3%	—	ImageNet (2D slices)
Ortiz et al. [73]	2022	3D CNN ensemble	PPMI	89%	0.87	Multiple pretraining

Table 5.9: Phase 1 PD detection performance compared with existing methods

Table 5.10 compares transfer learning strategies across medical imaging studies, validating our two-stage progressive unfreezing approach against established methodological recommendations in the literature.

Study	Year	Transfer Strategy	Source Domain	Target Task	Performance Gain
Our Work	2026	Two-stage (freeze→unfreeze)	ImageNet	PD cognitive decline	AUC: 0.76→0.8357 (+9.9%)
Tajbakhsh et al. [74]	2016	Fine-tuning vs scratch	ImageNet	Medical imaging (various)	10-30% improvement
Raghu et al. [75]	2019	Layer analysis	ImageNet	Medical imaging	Lower layers transfer well
Alzubaidi et al. [76]	2021	Staged fine-tuning	ImageNet	Medical imaging	Recommended progressive unfreezing
Ostertag et al. [61]	2023	Domain-specific transfer	ADNI (Alzheimer's)	PD cognitive decline	AUC: 0.72→0.81

Table 5.10: Transfer learning approaches in medical imaging and our implementation

Table 5.11 compares explainable AI implementations and neuroanatomical findings across studies, demonstrating that our attention-based interpretability identifies brain regions consistent with established PD dementia pathology.

Study	Year	XAI Method	Brain Regions Identified	Clinical Validation
Our Work	2026	Attention-based spatial maps	Frontal-parietal, prefrontal cortex, hippocampus, basal ganglia	Consistent with Gratwicke et al. (2015)
Beheshti & Ko [65]	2024	SHAP values	Frontal, occipital, cerebellum, midbrain	Validated against PD-MCI pathology
Shin et al. [77]	2021	Feature importance (bootstrap)	Angular gyrus, olfactory frontal areas	Radiology journal peer-reviewed
Forkert et al. [78]	2023	Saliency maps	Frontotemporal, orbital-frontal, deep gray matter	Multi-site validation
Lanskey et al. [79]	2020	Neuroimaging review	Frontal-parietal network critical for PD cognition	Reference for attention validation
Gratwicke et al. [80]	2015	Pathology review	Prefrontal, posterior parietal, hippocampus, basal ganglia	Gold standard for PD dementia regions

Table 5.11: XAI implementation and neuroanatomical validation across studies

5.6 Discussions

Why T1 Imaging Wins for Predicting Cognitive Decline?

T1-weighted imaging stands out in forecasting cognitive decline because of its exceptional gray-white matter contrast that enables the detection of cortical atrophy patterns that follow the progression of neurodegeneration well before dementia occurs. The frontal and parietal regions associated with PD-related cognitive decline show atrophy on T1 images well before the onset of dementia. The parallel convolution branches of Inception-V3 operating at multiple scales (1×1 , 3×3 , 5×5) are well suited to capture these changes. The attention-based interpretation method further verifies that the model indeed focuses on these frontal and parietal regions, which is consistent with neuroscientific studies that show that PD dementia involves networks that include these regions.

Why Clinical Fusion Underperformed?

The underperformance of Phase 2 fusion results can be explained by several reasons: (1) sample size: The sample size of 175 patients (122 patients in the training set) is not sufficiently large to learn complex interactions that require tuning many parameters; (2) feature-target alignment: The features used in clinical fusion are based on the present cognitive status and not on future cognitive decline, which makes it less suitable for predicting cognitive decline in 2-5 years; (3) prediction window: The prediction window might be beyond the time frame during which clinical trajectory features remain predictive; and (4) information redundancy: Motor symptoms, including PIGD, might be related to brain structure but add no new information beyond what MRI provides, which is essentially noise added to the model. Although this result is disappointing at first glance, it is actually a very important scientific contribution that will help guide future research directions.

Clinical Implications

The kicker here is that the best predictions are achieved by employing only T1-weighted imaging. This changes the way we operate because now we can determine the risk of cognitive impairment based solely on standard structural MRI scans, without needing additional T2 imaging or a series of tests. This streamlines implementation and could lower costs by as much as 40-50% compared to multi-modal approaches. With a 95% recall rate, most patients at risk are caught for closer monitoring and earlier interventions. The XAI attention maps provide a clear graphical representation for the clinician to interpret, allowing them to verify the AI's decision against their own clinical judgment and to communicate the risk evaluation with the patient.

Explainability and Clinical Trust

The attention-based explainability module addresses a major challenge in applying AI for dementia prediction, which is the black box problem. Since the model provides spatial attention maps that highlight the frontal cortex, temporal lobe, basal ganglia, and periventricular regions, which are all areas associated with cognitive decline in PD patients, it provides a rationale that is consistent with medical training. The quantitative attention measures also enable objective comparisons among patients and the ability to monitor changes in attention over time. This is very important for trust in a clinical context, particularly when AI predictions affect high-stakes clinical decisions and early dementia prevention plans.

Limitations

The sample size of the study (175 patients, 122 for training) limits the extent to which the model can be developed and the generalizability of its performance. External validation is required. The results may be population-specific, referring to the

PPMI population, which consists of PD patients enrolled in a research study. Clinical populations may vary in terms of disease severity and comorbidities as there was no subgroup analysis with respect to age, sex, and disease duration, which would help assess fairness with regard to different patient populations.

Chapter 6

Conclusion

6.1 Summary of Findings

The objectives of this research are to develop and rigorously evaluate deep learning architectures for PD detection and cognitive decline prediction, with the ultimate goal of facilitating early detection of dementia progression in PD patients using artificial intelligence.

This research is divided into two phases. Phase 1 of this study involves the development of a Parkinson’s disease detection model and the comparative analysis of the effectiveness of five different 3D CNN architectures: ResNet-18, DenseNet-121, EfficientNet-B0, VGG-16, and Inception-V3. The results show that the Inception-V3 architecture is more effective in PD detection than other architectures. The accuracy of the model is 78.53%, has an AUC-ROC of 0.8252, 0.8875 precision, recall is 0.8320, F1 score is 0.8589 and Matthews correlation coefficient is 0.4159. This demonstrates the model’s potential to utilize multi-scale feature extraction to detect neuroanatomical characteristics.

Phase 2 of this research is the main contribution of this research, as it addresses the biggest problem of predicting cognitive decline in PD patients who are at risk of dementia. The main contributions of this research are as follows: in the ablation study, it was observed that when only T1-weighted structural MRI is used, the model achieves its best results, and again, the Inception-V3 model outperforms all other architectures, achieving 0.8357 AUC, 85.19% accuracy, 86.4% precision, 95% recall, and 0.9048 F1-score, outperforming all other multimodal approaches. The recall of 95% is very high, making the model suitable for screening patients. On the other hand, the use of multimodal imaging did not show any improvement, where the use of T2 imaging did not show any improvement over the results of the T1-weighted approach, as the AUC of both approaches was the same, i.e., 0.8357. However, the inclusion of clinical features in the model resulted in a lower AUC of 0.7278 compared to the baseline, indicating that simpler models can achieve better results than the more complex approach. The proposed integrated attention-based explainable artificial intelligence mechanism results in the generation of spatial maps that identify the regions of the brain responsible for the predictions. It was observed that the model focuses its attention on the frontal cortex, temporal lobe, basal ganglia, and periventricular regions, which have been associated with PD-

related cognitive decline. It was also observed that the mean value of the attention in the patient population that declined cognitively was higher than that of the stable patient population. In addition, it was observed that 37.2% of the patient population did not show monotonic progression in the NP1COG scores, indicating that the approach of sustained progression, which requires the occurrence of two or more consecutive high visits, should be used. Finally, the approach of transfer learning from the ImageNet dataset was found to be effective in adapting to the task of volumetric 3D MRI analysis, where the second stage of transfer learning was found to be critical.

6.2 Contributions to the Field

This research makes several significant contributions to the field of deep learning-based prediction of cognitive decline in PD. First, the results of the Phase 1 experiment, which compares five different 3D CNN architectures, provide useful guidance in designing neuroimaging research, where it is established that Inception V3 is the best model for PD-related research due to its feature extraction capabilities. It gives the best result from the other architectures, like ResNet-18 which has more parameters and EfficientNet-B0 which has fewer parameters. Moreover, the ablation study, that was done in phase 2 compares seven different model configurations, namely T1-only, T2-only, T1+T2, Clinical-only, T1+Clinical, T2+Clinical, and T1+T2+Clinical, provides a much-needed framework of comparison that is largely lacking in existing research. Furthermore, the detailed discussion of negative results in terms of fusion degradation is an important contribution that might prevent researchers from developing overly complicated fusion architectures due to limited datasets. Third, the application of attention-based XAI is an important advancement in clinical AI research, particularly in the field of neurodegenerative diseases, where it is established that attention maps can be used to identify brain regions that contribute to cognitive decline prediction. Moreover, the results of this research, which show that attention maps correspond to established PD dementia neuroanatomy, such as frontal-parietal cortex, basal ganglia, and temporal structures, provide an important foundation for clinical trust and acceptance. From a clinical perspective, the results of the T1-only imaging providing the optimal predictive performance greatly facilitate the potential implementation of such an approach, allowing for the risk assessment of cognitive decline for early dementia detection using structural MRI scans without the need for T2 scans or extensive clinical testing. This would also potentially reduce the cost of such scans by approximately 40-50%. Moreover, the results would provide superior predictive accuracy, with an AUC of 0.8357 for T1-only images and an AUC of 0.7278 for T1+Clinical fusion images, and would be interpretable using XAI visualizations for validation by clinicians. The sustained progression method for addressing fluctuations due to NP1COG, affecting 37.2% of patients, provides a more accurate endpoint definition for PD cognitive decline studies, with potential applications for clinical trials where such an endpoint is critical for neuroprotective interventions.

6.3 Recommendations for Future Work

Immediate Extensions

The model should be externally validated on other Parkinson’s disease populations, such as those from the Parkinson’s Disease Biomarker Program (PDBP), Oxford Parkinson’s Disease Centre (OPDC), or other clinical populations. The implementation of GradCAM (Gradient-weighted Class Activation Mapping) will offer complementary gradient-based visualizations along with attention maps, thereby providing diverse views on model interpretability. The model performance should be assessed across demographic groups based on age, sex, disease duration, and genetic status (GBA and LRRK2 carriers) to ensure equal prediction performance across all patient groups and to identify potential biases that need to be addressed accordingly.

Multi-Modal Imaging Integration

Future studies should also include more imaging techniques to further elucidate the picture of neurodegenerative biomarkers, such as fMRI to investigate connectivity disruptions, DTI to measure white matter integrity, DaT SPECT or PET to measure nigrostriatal degeneration, amyloid PET, or tau PET to identify Alzheimer’s pathology, FDG PET to identify cortical hypometabolism, and neuromelanin-sensitive MRI to measure substantia nigra cell loss.

Advanced Architectural Approaches

The investigations should include CNN-LSTM or CNN-Transformer architectures that can model the longitudinal sequences of MRI scans, thereby capturing the trajectories of change rather than point predictions. The 4D video analysis of the scans could also help identify accelerations in the progression of atrophy that may signal impending cognitive decline. Brain connectivity graphs constructed from structural covariance or DTI can be analyzed with graph convolutional networks, which can capture changes in the brain networks beyond the conventional analysis of regional atrophy rates. Large-scale pre-training of the model on large quantities of unlabeled neuroimaging data can be done with contrastive or masked autoencoding objectives, which can help improve the quality of the feature representations before fine-tuning the model on the limited labeled data of patients with cognitive decline.

Final Remarks

As the field of Parkinson’s disease research continues to accumulate increasing numbers of longitudinal studies through collaborative research initiatives, the methodological framework that has been described in this paper will be valuable in the creation of increasingly sophisticated multi-modal approaches and will allow these tools to reach their maximum potential. While the integration of structural MRI with functional imaging, diffusion imaging, and molecular PET has the potential to provide a comprehensive imaging biomarker for the progression of neurodegeneration, this will require concerted effort in the form of data collection and validation to provide this benefit to the millions of people worldwide suffering from Parkinson’s disease, with the ultimate goal of identifying these individuals before the onset of the disease and within the therapeutic window in which prevention can provide a significant impact on the long-term outcome of the disease process.

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